LOST AND FOUND PORTAL – LOST AND FUND PORTAL SYSTEM

A PROJECT REPORT

Submitted by

TEJASWINI
(2414011114)

in partial fulfilment of the End Semester Examination for the course

CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF TECHNOLOGY COMPUTER SCIENCE AND BUSINESS SYSTEMS



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)
Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

2026–2027

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BONAFIDE CERTIFICATE

Certified that this project report titled "LOST AND FOUND PORTAL – CAMPUS ITEM MANAGEMENT SYSTEM USING ZOHO CREATOR AND ZIA" is the bonafide work of Tejaswini (2414011114), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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INTERNAL EXAMINER **EXTERNAL EXAMINER**

DECLARATION

I, Tejaswini (2414011114), hereby declare that the project report titled "LOST AND FOUND PORTAL – CAMPUS ITEM MANAGEMENT SYSTEM USING ZOHO CREATOR AND ZIA", submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report – including its data model, forms, reports, dashboard, Deluge scripts and workflows – was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. The AI-assisted feature Zia of Zoho Creator was

used to support the creation of the application, and this use is acknowledged in Section 5.1 and Section 5.9.

Place: Chennai
Date: _____

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ACKNOWLEDGEMENT

I express my sincere gratitude to our Chairman Mr. S. Meganathan, B.E., F.I.E., our Chairperson Dr. Thangam Meganathan, Ph.D., and our Vice-Chairman Mr. Abhay Shankar Meganathan for providing the infrastructure and facilities that made this work possible.

I am grateful to our Principal, Dr. S.N. Murugesan, M.E., Ph.D., Rajalakshmi Engineering College, for the constant encouragement extended to students to take up industry-relevant learning.

I thank Dr. K.R. Sowmia, Ph.D., Head of the Department, Department of Computer Science and Business Systems, for the guidance and support provided throughout the course.

I owe my sincere thanks to my course faculty, Dr. Domi Evangeline S, Assistant Professor (SG), for the valuable guidance, reviews and feedback that shaped this application and report.

I also acknowledge Zoho Corporation for providing access to the Zoho Creator platform and learning resources, and I thank my family and friends for their unwavering support.

TEJASWINI

ABSTRACT

Handling lost and found belongings on a college campus is still largely a manual activity. Students and staff misplace identity cards, books, wallets, keys and electronic devices, yet reporting depends on word of mouth, notice-board announcements and paper registers that give no permanent record, no dependable way of connecting a finder with an owner, and no evidence that a belonging was returned to the right person. This project presents the Lost and Found Portal, a campus item management application developed for Rajalakshmi Engineering College on the Zoho Creator low-code platform, with the Zia assistant supporting the creation of the application. The application is organised around ten related forms – User Registration, Lost Items, Found Items, Item Tracking, Claim Requests, Claim Verification, Item Handover, Item Status, Notifications and Administrator Management – connected through Lookup fields, so that user and item details are stored once and referred to wherever they are needed. A student or staff member reports a lost item, another user reports a found item, the administrator searches and matches the records, the owner submits a claim request, the claim is verified, the item is handed over and its status is updated. Sixteen reports and an administrative dashboard named LOST AND FOUND PORTAL – ADMIN DASHBOARD present four indicator cards, four horizontal bar charts, lists of recent activity and quick links to the forms. The report also specifies the proposed Delugedeployed validations, default-status rules and notification records, and states clearly which of these items remain to be verified in the application. Twenty-five test cases covering data entry, validation, lookup relationships, the dashboard and access control have been prepared; no execution result is claimed, and measured outcomes are to be recorded against them during evaluation. The present limitations are that matching, claim verification and notification depend on administrator action unless the proposed workflows are enabled, and that AI-based item matching, image recognition and external integrations are outside the current scope and are recorded as future enhancements.

Keywords: Low-Code Development; Zoho Creator; Zia; Lost and Found Management; Campus Administration; Lookup Relationships

PROJECT AT A GLANCE

Project Title	LOST AND FOUND PORTAL – Campus Item Management System using Zoho Creator and Zia
Application Domain	College Administration / Campus Item Management
Student Name & Register No.	Tejaswini – 2414011114

Institution	Rajalakshmi Engineering College (An Autonomous Institution, Affiliated to Anna University, Chennai)
Course	CB23F35 – Low-Code Application Development using Zoho Creator
Department / Semester / Year	Computer Science and Business Systems / V / 2026–2027
Project Type	Individual Low-Code Application Development Project
Application Name (Link Name)	Lost and Found Portal
Demo Credentials (per role)	Supplied separately at the time of demonstration
Number of Forms / Subforms	10 / 0 (User Registration, Lost Items, Found Items, Item Tracking, Claim Requests, Claim Verification, Item Handover, Item Status, Notifications, Administrator Management)
Number of Reports	16 in total (Table 5.1)
Number of Pages / Dashboards	1 (LOST AND FOUND PORTAL – ADMIN DASHBOARD)
Number of Workflows	Form workflows as listed in Table 5.3; no schedule, approval, Blueprint or batch workflows
Deluge Scripts	11 (DS-01 to DS-11, Table 5.2)
Integrations	No external API or third-party integration; email notification through the Deluge sendmail task is proposed
Roles & Profiles Defined	Administrator, Student, Faculty / Staff
AI Features	Zia used to assist application creation (design time). No operational AI feature (item matching, image recognition, classification) is claimed.
Development Environment	Zoho Creator web builder
Deployment Status	Developed on Zoho Creator; production deployment is not claimed

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
AI	Artificial Intelligence
API	Application Programming Interface
CO	Course Outcome
CSBS	Computer Science and Business Systems
DS	Deluge Script (script identifier)

ERD	Entity–Relationship Diagram
FR	Functional Requirement
HoD	Head of the Department
ID	Identifier / Identity
KPI	Key Performance Indicator
NFR	Non-Functional Requirement
QR	Quick Response (code)
RBAC	Role-Based Access Control
TC	Test Case
UAT	User Acceptance Testing
UC	Use Case
UI	User Interface
URL	Uniform Resource Locator
Zia	Zoho’s AI assistant

CHAPTER 1

INTRODUCTION

1.1 Background

A residential engineering college campus is a small town in which thousands of students, faculty members and staff move between classrooms, laboratories, libraries, hostels, canteens and sports grounds every day. In the course of this movement, belongings are routinely misplaced. The items that are most frequently lost include identity cards, books and notebooks, wallets, keys, water bottles, bags, umbrellas, spectacles, earphones, calculators, pen drives, mobile phones and other electronic devices. For a student, the loss of an identity card or a laboratory record can interrupt academic work, and the loss of a wallet or a mobile phone causes distress and expense. Rajalakshmi Engineering College, Chennai, is a large autonomous institution in which such losses occur throughout the academic year.

Items that are lost on a campus are usually found by another person within a short time, but the two parties rarely meet. A person who finds an item has no clear way of identifying its owner, and the owner has no clear way of learning that the item has been found. The finder may keep the item, leave it where it was found, or hand it to a security guard, a department office or a classroom teacher. The owner, meanwhile, asks friends, searches the places visited earlier, or enquires at several offices. Whether the item is recovered depends on chance, on the goodwill of individual staff members and on how quickly word spreads, rather than on a defined process.

The common mechanisms for dealing with such situations – verbal announcements in classrooms, notices on boards, messages in class groups and paper registers kept by an office – are informal and fragmented. They do not create one searchable record of what has been lost and what has been found, they do not record who claimed an item or how ownership was checked, and they leave no trace of whether an item was ever returned. Items that remain unclaimed are stored without an inventory and may eventually be misplaced themselves.

This project, titled **LOST AND FOUND PORTAL**, addresses this gap by building a centralised, web-based application on the Zoho Creator low-code platform, with Zia, the AI assistant of Zoho Creator, supporting the creation of the application. The portal records users, lost items, found items, possible matches, claim requests, claim verifications, item handovers, item status changes, notifications and administrators, and it presents the information through reports and an administrative dashboard. The

application domain is college administration, and the specific area is campus item management.

1.2 Problem Statement

Colleges lack a single, structured and traceable channel through which students and staff can report lost belongings, report items they have found, search the records, claim an item with proof of ownership, and have the return of the item recorded, while the administration monitors the whole activity. As a result, lost items are not reunited with their owners, found items are held without records, and claims cannot be checked or tracked. The specific pain points are listed below.

- Lost items are reported verbally, on notice boards or in messaging groups, so no permanent and searchable record of what was lost, where and when exists.
- A person who finds an item cannot easily identify the owner or the correct office, and the owner cannot easily learn that the item has been found.
- Found items are held informally by individuals or offices without an inventory, so the same item may be recorded in several places or in none.
- There is no uniform way of making a claim, of checking the claimant's proof of ownership or of recording that the item was handed over to the right person, which allows false claims and disputes.
- The status of an item – lost, found, under claim or returned – is not tracked, and the administration has no consolidated view of the number of items reported, held, claimed and returned.
- Contact details and information about valuable belongings are circulated in open groups, which exposes personal information.

1.3 Objectives

The objectives of the project are stated below. Each objective is traced to the implementation sections of Chapter 5 and to the results in Chapter 8 (Table 8.1).

- To design a data model of ten forms – User Registration, Lost Items, Found Items, Item Tracking, Claim Requests, Claim Verification, Item Handover, Item Status, Notifications and Administrator Management – connected through Lookup fields without repeating user and item details.
- To provide forms with which users report lost items and found items, capturing the item name, category, description, location, date and an optional image, with an automatically generated identifier and a default item status.

- To support searching, filtering and viewing of lost and found records through sixteen reports, and to record identified matches between lost and found items in the Item Tracking form.
- To manage the complete claim cycle through the Claim Requests, Claim Verification and Item Handover forms, so that every claim has a status and every returned item has a handover record.
- To provide an administrative dashboard named LOST AND FOUND PORTAL – ADMIN DASHBOARD with four indicator cards, four horizontal bar charts, lists of recent activity and quick links, all calculated from the stored records.
- To define validation and default-status rules, a role-based access design and a test plan of twenty-five traceable test cases, and to describe the deployment, security and governance of the application within the Zoho Creator environment.

1.4 Scope of the Project

The scope of the project is limited to the recording, searching, claiming and returning of belongings lost or found on the college campus. The items inside and outside the scope are as follows.

In scope:

- Registration of users – students, faculty members and staff – and of the administrators who manage the portal.
- Reporting of lost items and found items with category, description, location, date and optional image.
- Recording of possible matches between lost and found items, and tracking of the status of each item.
- Submission, verification and tracking of claim requests, and recording of the handover of the item to its owner.
- Notification records that inform the persons concerned about important events, and the LOST AND FOUND PORTAL – ADMIN DASHBOARD for monitoring.
- Three conceptual user groups – Administrator, Student and Faculty / Staff – accessed through the Zoho Creator web interface. Access through the Zoho Creator mobile app or a portal is considered in Section 5.11 and is subject to verification.

Out of scope (future enhancements):

- Automatic matching of lost and found items, image-based identification and QR-code labelling of items (Section 9.2).
- Integration with the college identity or student information system, and automatic reminders for unclaimed items.
- Payment, reward handling, courier or off-campus recovery of items.
- Custom pro-code components, which are discussed only as future work (Section
  5.10).

Throughout this report, features that are part of the project specification are described as implemented design, and every item whose existence or behaviour in the live application has not been confirmed is marked as to be verified. Future enhancements are never described as existing functionality.

1.5 Justification for a Low-Code Approach

Low-code application platforms allow business applications to be built through visual modelling, pre-built components and limited scripting, instead of fully hand-written code [1], [2]. For the Lost and Found Portal this approach is appropriate for several reasons. First, the application is a data-centred system made of forms, reports and status changes, which is the class of application that Zoho Creator supports through its form builder, report types and page builder [2]. Second, the development effort is small enough to be completed by a single student developer within a semester, and the resulting application can later be maintained by a citizen developer in the college administration without specialised programming skills. Third, Zoho Creator provides hosting, backup, user management, role-based security and access through browsers and mobile devices as built-in services, so that the project does not require separate infrastructure. Fourth, the Zia assistant can generate a first version of forms and fields from a description, which shortens the initial build, although every generated element has to be reviewed by the developer. Finally, the Deluge scripting language offers a controlled way of adding the few business rules that configuration alone cannot express, such as duplicate checks and automatic status changes. The trade-offs of the approach, including dependence on the platform vendor [1], are discussed in Section 8.4.

1.6 Organisation of the Report

Chapter 2 reviews the existing manual system and the related low-code literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design, including the data model, the form specifications, the process flows and the access control design. Chapter 5 details the implementation on Zoho Creator. Chapter 6

covers testing and debugging. Chapter 7 discusses deployment, performance optimisation, security and governance. Chapter 8 presents the results and discussion, and Chapter 9 concludes the report with future enhancements and learning outcomes. The references and Appendices A to F follow, containing the feature coverage checklist, the course outcome mapping, the Deluge source code, the user manual, the evaluation rubric and the viva-voce record sheet.

Reading note: where this report describes a proposed design, a planned script or an unverified outcome, the text says so explicitly. Items highlighted in yellow are placeholders that must be completed or verified by the student against the live Zoho Creator application before submission. Screenshot spaces are reserved for the student's own screenshots; no screenshot has been fabricated.

CHAPTER 2

LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

The existing arrangement for handling lost and found belongings in a typical college campus is manual and fragmented. The description below reflects this common pattern and should be confirmed against the practice actually observed at Rajalakshmi Engineering College.

A person who loses an item first tries to recall where it was last seen and searches those places. The person then asks friends and classmates, and may request the class teacher or a faculty member to announce the loss in the classroom or in the class messaging group. A notice may be pasted on a board. Another person who finds the item may keep it with himself or herself, hand it to a security guard or to the office of a department, or leave it on a desk. Where an office maintains a paper register, the item may be entered with a date and a brief description; in many cases no entry is made. The owner then visits several offices to ask whether the item has been found. When an item is shown to the owner, ownership is checked only by the personal judgement of the staff member present, and the handover is rarely recorded.

The as-is process flow proceeds in sequence as follows. Step 1: the person loses a belonging on the campus. Step 2: the person searches personally and asks friends. Step 3: an informal announcement is made in class, on a notice board or in a messaging group. Step 4: the finder keeps the item or hands it to staff, security or an office. Step 5: an entry is made in a paper register, if one is maintained. Step 6: the owner visits several offices to enquire. Step 7: the staff member checks ownership informally. Step 8: the item is handed over without a uniform record, or it remains unclaimed or is misplaced. The flow therefore has no recorded start, no tracked intermediate status and no defined end. Figure 2.1 depicts this flow.

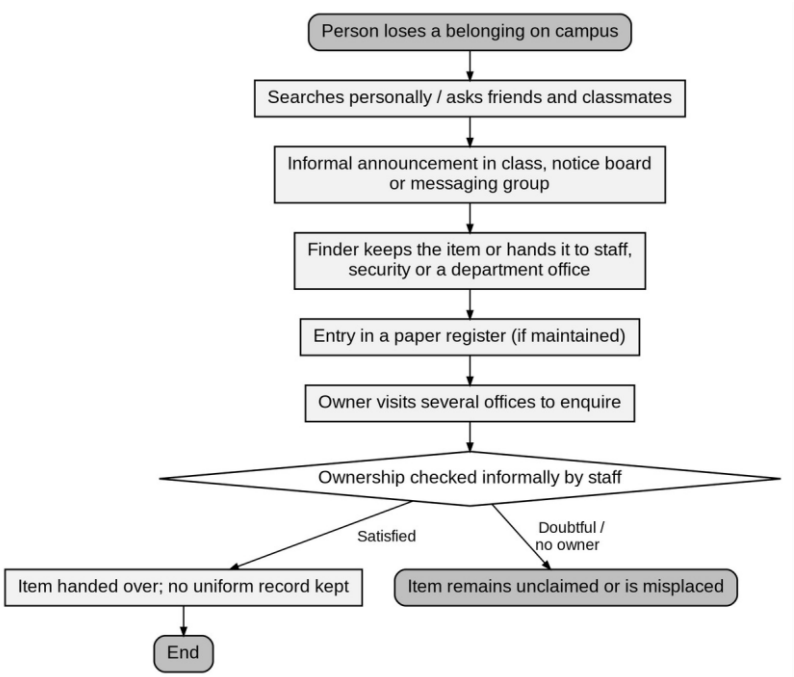


Figure 2.1: As-Is Process Flow of the Existing Manual Lost-and-Found System

2.2 Limitations of the Existing System

- Paper-based reporting: records in registers and on slips cannot be searched, sorted or shared, and they are available only at the place where they are kept.
- Manual announcements: notices and messages reach only the people who happen to see them, become outdated quickly and disclose contact details openly.
- Difficulty in identifying owners: a finder usually has no means of knowing whom an item belongs to, and an owner has no means of knowing where an item is held.
- No centralised records: lost reports and found reports are kept separately, if at all, so a lost item and its matching found item are never brought together systematically.
- Difficulty in tracking claims: a claim has no identifier, no status, no verifier and no recorded outcome, so disputes cannot be resolved from evidence.
- Duplicate and inconsistent information: the same item may be described differently in several places, and person details are written repeatedly.
- Delays in returning belongings: items are returned late or not at all because the right office or person is not known.
- No access control or audit: anyone who holds a register or a group message can read or alter it, and there is no history of who handled an item.

2.3 Low-Code versus Traditional Development

Table 2.1 compares traditional pro-code development with low-code development on Zoho Creator for the Lost and Found Portal. The traditional-development figures are planning estimates for an application of this size and are not measurements. The lowcode column records the effort that the student must enter from the project log. General discussions of the comparison are available in [1] and [2].

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	Estimated 8–10 weeks for design, database, back end, front end and testing (planning estimate).	Estimated 2–3 weeks for ten forms, reports and a dashboard (planning estimate).
Skills required	Web front end, server-side language, relational database design, authentication and deployment skills.	Data modelling, form and report configuration, and basic Deluge scripting; Zia assists the first draft of forms.
Hosting and infrastructure	Server or cloud hosting, database administration, backups and security patches to be arranged.	Hosting, database, backups and platform security are provided by Zoho Creator.
Search and filtering	Search pages, queries and filters must be coded and tested.	Report search, filters and grouping are configured in the report builder.
Mobile application	A separate mobile application or a responsive front end is required.	Applications are accessible through the Zoho Creator mobile app without separate development.
Security and access control	Authentication, authorisation and session management must be implemented and tested.	Built-in users, roles, permission profiles and record-level criteria.
Maintenance and change	Changes require code modification, rebuilding and redeployment by a developer.	Forms, fields and reports can be modified by a citizen developer in the builder.
Customisation limits	Virtually unlimited, at the cost of effort.	Limited to platform features, with Deluge and integrations as extension points.
Cost	Developer time and hosting costs; no licence fee for open-source stacks.	Platform subscription or education access; very low development and hosting cost.

2.4 Literature Survey

Table 2.2 summarises the sources reviewed. They comprise peer-reviewed articles on low-code platforms and the official Zoho documentation. The bibliographic details are given in the References section and must be re-verified against the original

publications before submission; entries whose details could not be confirmed are marked for verification.

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	A. C. Bock and U. Frank, 2021 [1]	Low-Code Platform	Characterises low-code platforms and discusses their benefits and limitations, including dependence on the platform vendor.	Supports the justification in Section 1.5 and the limitations in Section 8.4.
2	Zoho Corporation [2]	Zoho Creator Help Documentation	Documents forms, fields, reports, pages, workflows, users and permissions, and deployment in Zoho Creator.	Primary technical basis for the design and implementation of the Lost and Found Portal.
3	R. Sanchis, Ó. GarcíaPerales, F. Fraile and R. Poler, 2020 [3]	Low-Code as Enabler of Digital Transformation in Manufacturing Industry	Discusses low-code as a means of faster development of business applications by users close to the process.	Supports the choice of low-code for a campus process maintained by non-specialists.
4	Zoho Corporation [4]	Deluge Script Reference Guide	Describes Deluge syntax, form events, record fetch, insert and update, and notifications.	Basis of the validation, default-status and notification scripts of Section 5.5.
5	D. Di Ruscio, D. Kolovos, J. de Lara, A. Pierantonio, M. Tisi and M. Wimmer, 2022 [5]	Low-code development and model-driven engineering: Two sides of the same coin?	Examines the relationship between low-code platforms and model-driven engineering and identifies open challenges.	Relates to the model-first design of the ten-form data model in Chapter 4.
6	M. Waszkowski, 2019 [6]	Low-code platform for automating business processes in manufacturing	Reports the use of a lowcode platform to automate business processes in a manufacturing context.	Supports status-driven automation of claim and item status changes.

The literature reviewed above concerns low-code platforms and their use for business processes. Authentic published studies on campus lost-and-found systems and on digital information management in college administration were not available to the student in verified form while preparing this report, and none has therefore been cited, so as to avoid unverified references. A survey of existing commercial or open-source lost-and-found products is also not included for the same reason.

2.5 Emerging Trends in Low-Code Development

Several trends in low-code development are relevant to the Lost and Found Portal.

Analyst publications on low-code application platforms, such as the Gartner Magic Quadrant [7], describe a continuing shift toward platforms that combine visual development, integration and artificial intelligence. Citizen development is the first trend: staff who understand the process but are not professional programmers build and maintain applications themselves [3]. The portal follows this trend by keeping its data model readable, so that an administrator can later add an item category or a status value without developer support.

The second trend is AI-assisted development and AI-enabled applications. Zoho Creator offers an assistant, Zia, which was used in this project at design time to support the creation of the application. AI-enabled operation – for example, suggesting a match between a lost description and a found description, or recognising an object in an uploaded photograph – is a different matter, and it is recorded as a future enhancement in Section 9.2 rather than as an implemented feature. The third trend is hyperautomation, in which workflows and notifications reduce manual hand-offs; the proposed status updates and notification records apply this idea at a modest scale. The fourth trend is composability, in which small applications exchange data through APIs [5]; the separation of users, items, claims and handovers into related forms keeps the portal open to later integration with the college identity system.

2.6 Proposed System

The proposed system is the Lost and Found Portal on Zoho Creator. It replaces informal channels with ten related forms and gives each stakeholder a defined view of the data. The way in which the proposed system overcomes each limitation listed in Section 2.2 is summarised below.

- **Paper-based reporting** is overcome by the Lost Items and Found Items forms, which store every report as a record with an automatically generated identifier, category, description, location, date and optional image.
- **Manual announcements** are replaced by searchable reports, so that any user can look for an item at any time, and by notification records that inform the persons concerned (Section 5.6).
- **Difficulty in identifying owners** is addressed by the Item Tracking form, which records the possible match between a lost item and a found item, and by the Claim Requests form, in which the claimant describes the item and its identifying marks.
- **No centralised records** is overcome by storing all users, items, claims, verifications and handovers in one application and linking them through Lookup fields.

- **Difficulty in tracking claims** is overcome by the Claim Status field, the Claim Verification form with its method, result and remarks, and the Item Handover form that closes each claim.
- **Duplicate information** is reduced because user details are entered once in User Registration and are referred to by Lookup fields in every other form.
- **Delays in return** are reduced because the Pending Claims report and the dashboard show the administrator which claims are waiting for action.
- **No access control or audit** is addressed by the role-based access design (Sections 4.6 and 5.8) and by the platform audit facilities (Section 7.4).

CHAPTER 3

REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Three conceptual stakeholder groups are identified for the application. Table 3.1 describes each group and its responsibilities. The groups describe the intended design; the configuration of corresponding Zoho Creator users and permission profiles is described, and its verification status stated, in Section 5.8. No custom role or permission profile is claimed to exist merely because a stakeholder group is described here. **Table 3.1: Stakeholders and User Roles**

Stakeholder	Description	Key Responsibilities in the Application
Administrator	College staff member (for example, from the security office or the administrative office) who manages lost and found activities and the application.	Manage user and administrator records; review lost and found records; record possible matches; verify claims; record handovers; update item status; maintain notification records; monitor reports and the dashboard.
Student	Registered student of the college who may lose an item, find an item or claim an item.	Register; report lost items; report found items; search and view items; submit claim requests; track own claims; view own notifications.
Faculty / Staff	Faculty member or non-teaching staff member of the college who may lose, find or claim an item, or hand a found item to the administrator.	Same functions as a student for reporting, searching and claiming; may also report items handed in by others.

3.2 Functional Requirements

The functional requirements are listed in Table 3.2. Each requirement is testable and is referred to by its FR identifier in the design tables of Chapter 4, the implementation tables of Chapter 5 and the test cases of Table 6.1. Requirements that depend on proposed scripts or workflows are marked “(proposed)”.deployed

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall allow a user to be registered with User Name, Register Number / ID, User Type, Email, Phone Number, Department, Year and Registration Date, with the mandatory details enforced.	Administrator, Student, Faculty / Staff	High
FR-02	The system shall prevent two users from being registered with the same Register Number / ID (proposed script DS-deployed 01).	Administrator, Student, Faculty / Staff	High

FR-03	The system shall allow a user to report a lost item with Item Name, Category, Description, Lost Location and Lost Date, linked to the reporting user through a Lookup field, and shall generate a Lost Item ID automatically.	Student, Faculty / Staff, Administrator	High

FR ID	Requirement	Role(s)	Priority
FR-04	The system shall reject a Lost Date that is later than the current date and shall set the default Item Status to Lost (proposed scripts DS-02, DSdeployed -03).	Student, Faculty / Staff	Medium
FR-05	The system shall allow a user to report a found item with Item Name, Category, Description, Found Location, Found Date and Storage Location, linked to the reporting user, and shall generate a Found Item ID automatically.	Student, Faculty / Staff, Administrator	High
FR-06	The system shall reject a Found Date later than the current date and shall set the default Item Status to Found (proposed scripts DS-04, DSdeployed -05).	Student, Faculty / Staff	Medium
FR-07	The system shall allow an optional image of a lost or found item to be uploaded and stored with the record.	Student, Faculty / Staff	Low
FR-08	The system shall allow lost and found items to be searched, filtered and viewed by category, location, date and status through reports.	All stakeholders (as permitted)	High
FR-09	The system shall allow the administrator to record a possible match between a Lost Item and a Found Item in the Item Tracking form with a Match Status and remarks.	Administrator	High
FR-10	The system shall allow a user to submit a claim request for a found item with an ownership description, identifying marks and an optional proof document, linked to the Found Item and the Claimant.	Student, Faculty / Staff	High
FR-11	The system shall set the default Claim Status to Pending and shall prevent a duplicate pending claim by the same claimant on the same found item (proposed scripts DS-06, deployed DS-07).	Student, Faculty / Staff	High
FR-12	The system shall allow the administrator to record the verification of a claim with Verification Method, Verification Result and remarks, and the Claim Status shall follow the result (proposed script DS-08).deployed	Administrator	High
FR-13	The system shall allow the administrator to record the handover of an approved claim with Handed Over By, Handover Date, ID proof check and remarks, and the related item statuses shall change to Returned (proposed deployed script DS-09).	Administrator	High

FR-14	The system shall maintain an item status record that shows the current status of a lost or found item, who updated it and when.	Administrator	Medium
FR-15	The system shall store notification records for important events and shall allow a recipient to view the notifications addressed to himself or herself (proposed script DS-10; deployed email delivery proposed).	All stakeholders (as permitted)	Medium
FR-16	The system shall allow administrator details to be recorded and shall refer to them in tracking, verification, handover and status records.	Administrator	Medium
FR-17	The system shall provide the sixteen reports listed in Table 5.1.	All stakeholders (as permitted)	Medium
FR ID	**Requirement**	**Role(s)**	**Priority**
FR-18	The system shall provide the LOST AND FOUND PORTAL – ADMIN DASHBOARD showing Total Lost Items, Total Found Items, Pending Claims and Successfully Returned Items, four charts, lists of recent activity and quick links, showing zero where no records exist.	Administrator	Medium
FR-19	The system shall let a student or staff member view and track only the records that relate to himself or herself, and shall let the administrator manage all records.	All stakeholders	High
FR-20	The system shall protect contact details and claim information so that they are visible only to permitted stakeholders.	All stakeholders	High

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Reports with up to 1,000 records shall load within 5 seconds under normal network conditions (design target; to be measured and recorded in Table 7.2).
NFR-02	Security and privacy	Users shall access only the data permitted by their role; a student shall not view the records, contact details or claims of another user.
NFR-03	Usability	The application shall be usable in a web browser and, where verified, in the Zoho Creator mobile app, with forms that a first-time user can complete without training.
NFR-04	Availability	The application shall be available whenever the Zoho Creator service is available, since hosting is provided by the platform.
NFR-05	Maintainability	Lists of values such as categories, user types, statuses and verification methods shall be modifiable in the form builder without scripting changes.

NFR-06	Auditability	Changes to records shall be traceable through the platform audit trail and the Added User and Added Time information of each record.
NFR-07	Data integrity	Relationships between forms shall be maintained through Lookup fields so that every item, claim, verification, handover, status entry and notification refers to an existing user, item, claim or administrator.
NFR-08	Data accuracy	Mandatory fields, date checks and drop-down lists shall be used so that item records are complete and consistently described.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	**Specification**
Platform	Zoho Creator
AI assistant	Zia (Zoho Creator) for AI-assisted application creation
Category	**Specification**
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome, Microsoft Edge or Mozilla Firefox (latest version)
Other Zoho services	Email delivery through the Deluge sendmail task (proposed); no other Zoho service is claimed
Mobile	Zoho Creator mobile app for Android or iOS
Client hardware	Intel Core i3 / i5 or above, 8 GB RAM, internet connection

3.5 Use Case Model

The use case model identifies two actor groups – the Student / Faculty / Staff user and the Administrator – and the main functions available to each. Figure 3.1 shows the use case diagram, and Tables 3.5 to 3.7 describe three key use cases in detail.

The user actor is associated with the use cases Register as User, Report Lost Item, Report Found Item, Search and View Items, Submit Claim Request, Track Claim Status and View Notifications. The administrator actor is associated with Manage Users, Manage Administrator Details, Review Lost and Found Records, Record Possible Match, Verify Claim Request, Record Item Handover, Update Item Status, Manage Notifications, and View Dashboard and Reports. The use case Submit Claim Request includes Search and View Items, because a claimant first finds the item in the records. The use case Record Item Handover depends on the use case Verify Claim Request, because an item is handed over only after a claim has been approved. Together these use cases form the item life cycle shared by the user and the administrator.



Figure 3.1: Use Case Diagram of the Application Table 3.5: Use Case Description – Report Lost Item

Use Case ID	UC-01
Use Case Name	Report Lost Item
Actor(s)	Student, Faculty / Staff
Pre-condition	The user is registered in the User Registration form and is logged in.
Main Flow	1. The user opens the Lost Items form. 2. The user selects the Reported By Lookup. 3. The user selects the Category and enters Item Name, Item Description, Lost Location and Lost Date, and optionally uploads an image. 4. The user submits the form; the system validates the entries, generates the Lost Item ID and sets the Item Status to Lost. 5. The system shows the success message.
Alternate Flow	If a mandatory field is empty or the Lost Date is later than today, the system shows a validation message and does not save the record.
Post-condition	A lost item record with status Lost exists and appears in the Lost Items Report and in the Total Lost Items card.
Related FR	FR-03, FR-04, FR-07

Table 3.6: Use Case Description – Report Found Item and Record Possible Match

Use Case ID	UC-02
Use Case Name	Report Found Item and Record Possible Match

Actor(s)	Student, Faculty / Staff, Administrator
Pre-condition	The finder is registered and logged in; the administrator is registered in Administrator Management.
Main Flow	1. The finder opens the Found Items form and selects Reported By. 2. The finder enters Category, Item Name, Description, Found Location, Found Date and Storage Location. 3. On submission the system generates the Found Item ID and sets the Item Status to Found. 4. The administrator searches the Lost Items Report for a similar record. 5. If a possible owner is found, the administrator opens the Item Tracking form, selects the Lost Item and the Found Item, selects Tracked By and sets Match Status to Possible Match.
Alternate Flow	If no similar lost item exists, the found item stays in storage with status Found until a claim is submitted.
Post-condition	A found item record exists; where a match was identified, an Item Tracking record links the two items.
Related FR	FR-05, FR-06, FR-08, FR-09

Table 3.7: Use Case Description – Claim, Verify and Return an Item

Use Case ID	UC-03
Use Case Name	Claim, Verify and Return an Item
Actor(s)	Student, Faculty / Staff, Administrator
Pre-condition	The found item exists with status Found; the claimant is registered and logged in.
Main Flow	1. The claimant opens the Claim Requests form and selects the Found Item and Claimant. 2. The claimant enters the ownership description and identifying marks and, optionally, uploads a proof document. 3. The system sets the Claim Status to Pending. 4. The administrator opens the Pending Claims report and then the Claim Verification form, and records the Verification Method, Result and remarks. 5. For an approved claim, the administrator opens the Item Handover form, selects the verification record, enters Handed Over By, Handover Date and ID Proof Checked, and submits. 6. The item statuses change to Returned and the dashboard reflects the new counts.
Alternate Flow	If the verification result is Rejected, the claim status becomes Rejected and the found item remains in storage with status Found. A duplicate pending claim by the same claimant on the same item is refused.
Post-condition	The claim has a verification record and, if approved, a handover record; the item status is Returned and the claim is completed.
Related FR	FR-10, FR-11, FR-12, FR-13, FR-14, FR-15

CHAPTER 4

SYSTEM DESIGN

4.1 System Architecture

The Lost and Found Portal follows the layered structure of a Zoho Creator application. Figure 4.1 shows the layers, and the explanation below states the role of each layer.

The architecture has five layers arranged from the user to the platform services. The access layer consists of the three stakeholder groups – Administrator, Student and Faculty / Staff – who reach the application through a web browser and, where verified, the Zoho Creator mobile app. The presentation layer consists of the application menu, the ten forms, the sixteen reports and the LOST AND FOUND PORTAL – ADMIN DASHBOARD. The logic layer consists of field settings and validations, the proposeddeployed Deluge scripts and the proposed form workflows that enforce business rules and set default values. The data layer consists of the records of the ten forms, connected through Lookup fields, and the system fields Added User and Added Time that the platform adds to every record. The services layer consists of the Zoho Creator hosting, the Zia assistant that supported the creation of the application at design time, and the proposed email notification through the sendmail task; no external API integration is part of the application. Role-based permission profiles cut across the presentation, logic and data layers and decide which components and records each user can reach.

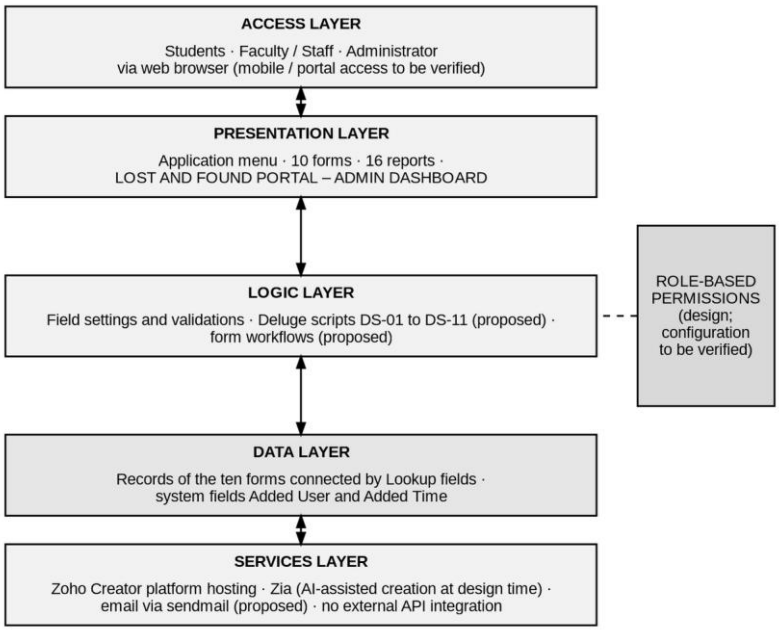


Figure 4.1: System Architecture of the Application

The access layer separates users by role, so that the same application can show different menus and records to an administrator and to a student. The presentation

layer lets users enter and view data without writing code. The logic layer contains the Deluge scripts described in Section 5.5; Deluge is the scripting language of Zoho Creator and runs on form events such as On Load, On Validate and On Success [4]. The data layer is managed by the platform, so that no database administration is needed. The distinction between design-time AI assistance and run-time AI functionality is kept explicit: the architecture contains no run-time AI component.

4.2 Data Model

The data model contains ten entities, each implemented as a Zoho Creator form. User Registration and Administrator Management are the two master forms for people. Lost Items and Found Items are the two master forms for belongings. Item Tracking, Claim Requests, Claim Verification, Item Handover, Item Status and Notifications are the transaction forms that refer to the master forms, or to one another, through Lookup fields. In Zoho Creator a Lookup field stores a reference to a record of another form and thereby realises a one-to-many relationship between the parent form and the child form. No Subform is used, because none of the ten forms contains a repeating group of rows that belongs to a single parent record.

The entity–relationship structure is described as follows. User Registration is related to Lost Items, Found Items, Claim Requests and Notifications: one user can report many lost items and many found items, can make many claims and can receive many notifications, and each such record refers to exactly one user. Lost Items and Found Items are related to Item Tracking, Claim Requests, Item Status and Notifications, so that each possible match, claim, status entry and notification can be traced to the item concerned. Claim Requests is related to Claim Verification, and Claim Verification is related to Item Handover, so that every handover is traceable to a verification and, through it, to a claim. Administrator Management is related to Item Tracking, Claim Verification, Item Handover and Item Status, identifying the administrator who performed each action. There are no many-to-many relationships between the forms. Figure 4.2 shows the diagram, in which 1 denotes the parent form and N the child form.

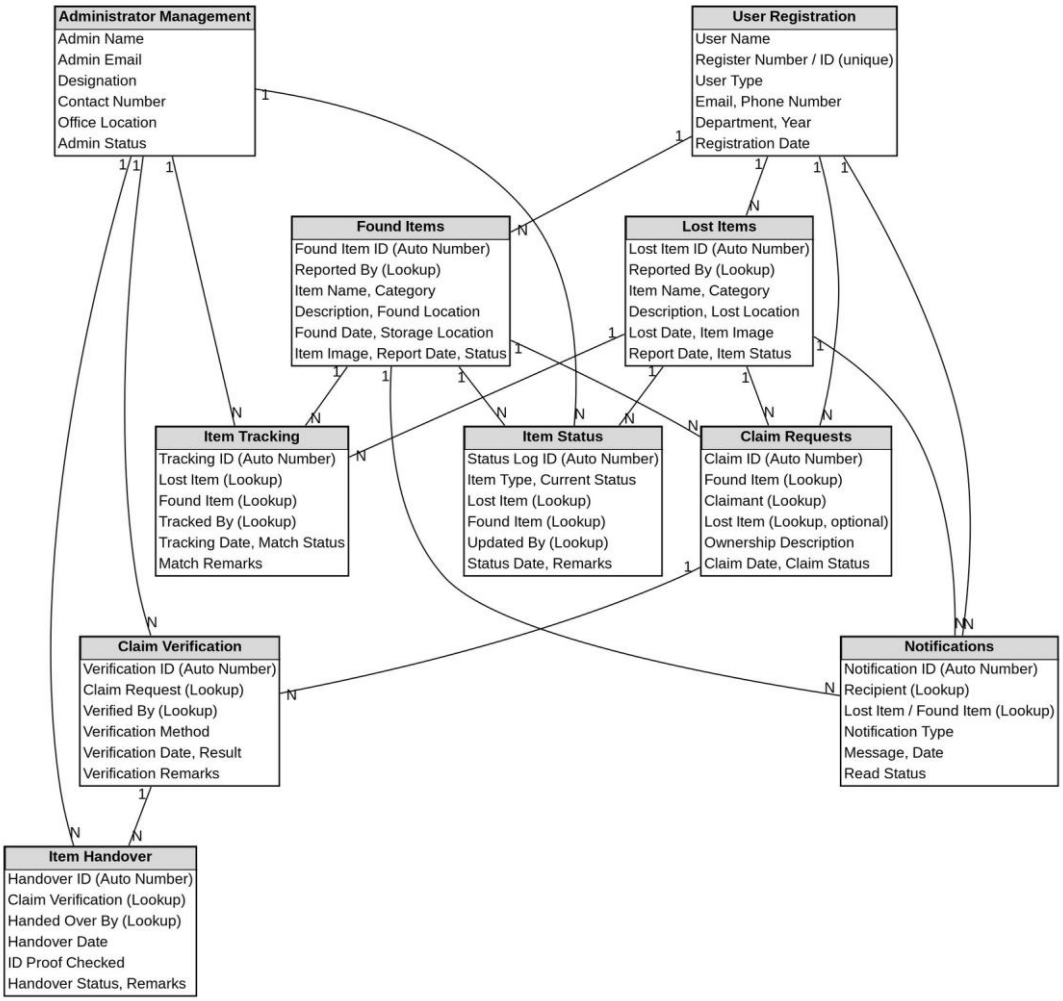


Figure 4.2: Entity–Relationship Diagram of the Ten Forms

Table 4.1 lists the Lookup relationships. Only the relationships that belong to the specified application are shown; no additional entity has been introduced. A relationship is a specified design until it is confirmed in the form builder, and each row is therefore marked for verification. The relationship between a lost item and a found item is deliberately indirect: it is recorded through the Item Tracking form (as a possible match) and through the optional Lost Item Lookup of a claim request, and not by a permanent link in the item forms themselves, because a match is a judgement that may later be changed.

Table 4.1: Entities and Relationships

Child Form	Lookup Field	Parent Form	Relationship Type	Status
Lost Items	Reported By	User Registration	Many-to-One	Specified
Found Items	Reported By	User Registration	Many-to-One	Specified

Item Tracking	Lost Item	Lost Items	Many-to-One	Specified
Item Tracking	Found Item	Found Items	Many-to-One	Specified
Child Form	**Lookup Field**	**Parent Form**	**Relationship Type**	**Status**
Item Tracking	Tracked By	Administrator Management	Many-to-One	Specified
Claim Requests	Found Item	Found Items	Many-to-One	Specified
Claim Requests	Claimant	User Registration	Many-to-One	Specified
Claim Requests	Lost Item (optional)	Lost Items	Many-to-One	Specified
Claim Verification	Claim Request	Claim Requests	Many-to-One	Specified
Claim Verification	Verified By	Administrator Management	Many-to-One	Specified
Item Handover	Claim Verification	Claim Verification	Many-to-One	Specified
Item Handover	Handed Over By	Administrator Management	Many-to-One	Specified
Item Status	Lost Item (conditional)	Lost Items	Many-to-One	Specified
Item Status	Found Item (conditional)	Found Items	Many-to-One	Specified
Item Status	Updated By	Administrator Management	Many-to-One	Specified
Notifications	Recipient	User Registration	Many-to-One	Specified
Notifications	Related Lost Item	Lost Items	Many-to-One	Specified
Notifications	Related Found Item	Found Items	Many-to-One	Specified

4.3 Form Design

This section gives the field specification of each of the ten forms in Tables 4.2 to 4.11. The functional areas of the portal map to the forms as follows: User Management – User Registration and Administrator Management; Lost Item Management – Lost Items;

Found Item Management – Found Items; Item Search and Tracking – Item Tracking (together with the search and filter facility of the reports); Claim Request Management – Claim Requests; Claim Verification – Claim Verification; Item Return / Handover Management – Item Handover; Item Status Management – Item Status; Notification

Management – Notifications. Administrative Management is supported by Administrator Management and by the administrative dashboard. The functional areas are therefore not one-to-one with the forms.

The field labels, link names, field types, mandatory settings, validations and drop-down values in the tables are the proposed design specification. They follow the default Zoho Creator convention in which the link name is the label with underscores and must be verified against the form builder before submission. Zoho Creator also adds the system fields Added User and Added Time to every record. Auto Number fields are systemgenerated, and the identifier prefix is to be confirmed.

Table 4.2: Field Specification – User Registration Form

Field Label	Proposed Link Name	Field Type	Mandatory (design)	Validation / Default	Remarks
User Name	User_Name	Single Line	Yes	Not empty	Full name of the student, faculty or staff member
Register Number / ID	Register_Number_ID	Single Line	Yes	No duplicate values (DS-01)	Unique identifier of the user
User Type	User_Type	Drop Down	Yes	Student / Faculty / Staff	Stakeholder group of the user
Email	Email	Email	Yes	Valid email format	Contact address; used for record-level access
Phone Number	Phone_Number	Phone	Yes	Valid phone format	Contact number (sensitive)
Department	Department	Drop Down	Yes	Choice list	Department of study or work
Year of Study	Year_of_Study	Drop Down	No	I / II / III / IV / Not applicable	Applicable to students only
Registration Date	Registration_Date	Date	No	Default: current date	Date of registration

Table 4.2 lists the proposed specification of the User Registration form (link name User_Registration). Link names, mandatory settings, validations and drop-down values are to be verified in Zoho Creator.

Table 4.3: Field Specification – Lost Items Form

Field Label	Proposed Link Name	Field Type	Mandatory (design)	Validation / Default	Remarks

Lost Item ID	Lost_Item_ID	Auto Number	System	Auto-generated	Identifier of each lost report
Reported By	Reported_By	Lookup (User Registration)	Yes	Selects an existing user	Relationship to User Registration
Item Name	Item_Name	Single Line	Yes	Not empty	Short name of the lost item
Category	Category	Drop Down	Yes	Identity Card / Books and Notebooks / Wallet / Keys / Electronic Device / Bag / Accessories / Clothing / Other	Used by the Lost Items by Category report and chart
Field Label	**Proposed Link Name**	**Field Type**	**Mandatory (design)**	**Validation / Default**	**Remarks**
Item Description	Item_Description	Multi Line	Yes	Not empty	Colour, brand, size, marks and contents
Lost Location	Lost_Location	Single Line	Yes	Not empty	Place on campus where the item was last seen
Lost Date	Lost_Date	Date	Yes	Not later than today (DS-03)	Date on which the item was lost
Item Image	Item_Image	Image	No	Optional	Photograph of the item, if available
Report Date	Report_Date	Date	No	Default: current date (DS-02)	Date of reporting
Item Status	Item_Status	Drop Down	Yes	Default: Lost (DS02); Lost / Possible Match / Returned / Closed	Drives reports, chart and dashboard

Table 4.3 lists the proposed specification of the Lost Items form (link name Lost_Items). Link names, mandatory settings, validations and drop-down values are to be verified in Zoho Creator.

Table 4.4: Field Specification – Found Items Form

Field Label	Proposed Link Name	Field Type	Mandatory (design)	Validation / Default	Remarks
Found Item ID	Found_Item_ID	Auto Number	System	Auto-generated	Identifier of each found report
Reported By	Reported_By	Lookup (User Registration)	Yes	Selects an existing user	Finder of the item

Item Name	Item_Name	Single Line	Yes	Not empty	Short name of the found item
Category	Category	Drop Down	Yes	Identity Card / Books and Notebooks / Wallet / Keys / Electronic Device / Bag / Accessories / Clothing / Other	Same category list as Lost Items
Item Description	Item_Description	Multi Line	Yes	Not empty	Visible description of the item
Found Location	Found_Location	Single Line	Yes	Not empty	Place on campus where the item was found
Found Date	Found_Date	Date	Yes	Not later than today (DS-05)	Date on which the item was found
Field Label	**Proposed Link Name**	**Field Type**	**Mandatory (design)**	**Validation / Default**	**Remarks**
Storage Location	Storage_Location	Single Line	Yes	Not empty	Place where the item is currently kept
Item Image	Item_Image	Image	No	Optional	Photograph of the item, if available
Report Date	Report_Date	Date	No	Default: current date (DS-04)	Date of reporting
Item Status	Item_Status	Drop Down	Yes	Default: Found (DS-04); Found / Under Claim / Returned / Closed	Drives reports, chart and dashboard

Table 4.4 lists the proposed specification of the Found Items form (link name Found_Items). Link names, mandatory settings, validations and drop-down values are to be verified in Zoho Creator.

Table 4.5: Field Specification – Item Tracking Form

Field Label	Proposed Link Name	Field Type	Mandatory (design)	Validation / Default	Remarks
Tracking ID	Tracking_ID	Auto Number	System	Auto-generated	Identifier of the tracking record
Lost Item	Lost_Item	Lookup (Lost Items)	Yes	Selects an existing lost item	Relationship to Lost Items
Found Item	Found_Item	Lookup (Found Items)	Yes	Selects an existing found item	Relationship to Found Items

Tracked By	Tracked_By	Lookup (Administrator Management)	Yes	Selects an existing administrator	Administrator who identified the match
Tracking Date	Tracking_Date	Date-Time	No	Default: current datetime	When the match was recorded
Match Status	Match_Status	Drop Down	Yes	Possible Match / Confirmed Match / Not a Match	State of the match
Match Remarks	Match_Remarks	Multi Line	No	—	Basis on which the items were matched

Table 4.5 lists the proposed specification of the Item Tracking form (link name Item_Tracking). Link names, mandatory settings, validations and drop-down values are to be verified in Zoho Creator.

Table 4.6: Field Specification – Claim Requests Form

Field Label	Proposed Link Name	Field Type	Mandatory (design)	Validation / Default	Remarks
Claim ID	Claim_ID	Auto Number	System	Auto-generated	Identifier of each claim
Found Item	Found_Item	Lookup (Found Items)	Yes	Selects an existing found item	Item being claimed
Claimant	Claimant	Lookup (User Registration)	Yes	Selects an existing user	Person making the claim
Lost Item	Lost_Item	Lookup (Lost Items)	No	Optional	Claimant's own lost report, if any
Ownership Description	Ownership_Description	Multi Line	Yes	Not empty (DS-06)	Details that only the owner would know
Identifying Marks	Identifying_Marks	Multi Line	No	—	Marks, stickers, contents, serial numbers
Proof Document	Proof_Document	File Upload	No	Optional	Bill, photograph or other proof
Claim Date	Claim_Date	Date	No	Default: current date (DS-06)	Date of the claim
Claim Status	Claim_Status	Drop Down	Yes	Default: Pending (DS06); Pending / Under Verification / Approved / Rejected / Completed	Drives Pending Claims report and dashboard card

Table 4.6 lists the proposed specification of the Claim Requests form (link name Claim_Requests). Link names, mandatory settings, validations and drop-down values are to be verified in Zoho Creator.

Table 4.7: Field Specification – Claim Verification Form

Field Label	Proposed Link Name	Field Type	Mandatory (design)	Validation / Default	Remarks
Verification ID	Verification_ID	Auto Number	System	Auto-generated	Identifier of the verification record
Claim Request	Claim_Request	Lookup (Claim Requests)	Yes	Selects an existing claim	Relationship to Claim Requests
Verified By	Verified_By	Lookup (Administrator Management)	Yes	Selects an existing administrator	Administrator who verified the claim
Verification Date	Verification_Date	Date	No	Default: current date	Date of verification
Field Label	**Proposed Link Name**	**Field Type**	**Mandatory (design)**	**Validation / Default**	**Remarks**
Verification Method	Verification_Method	Drop Down	Yes	ID Card Check / Description Match / Photo Comparison / Other	How ownership was checked
Verification Result	Verification_Result	Drop Down	Yes	Approved / Rejected	Outcome used by DS-08
Verification Remarks	Verification_Remarks	Multi Line	No	—	Reason or notes of the administrator

Table 4.7 lists the proposed specification of the Claim Verification form (link name Claim_Verification). Link names, mandatory settings, validations and drop-down values are to be verified in Zoho Creator.

Table 4.8: Field Specification – Item Handover Form

Field Label	Proposed Link Name	Field Type	Mandatory (design)	Validation / Default	Remarks
Handover ID	Handover_ID	Auto Number	System	Auto-generated	Identifier of the handover record
Claim Verification	Claim_Verification	Lookup (Claim Verification)	Yes	Selects an approved verification	Relationship to Claim Verification
Handed Over By	Handed_Over_By	Lookup (Administrator Management)	Yes	Selects an existing administrator	Administrator who released the item

Handover Date	Handover_Date	Date-Time	Yes	Default: current datetime	When the item was handed over
ID Proof Checked	ID_Proof_Checked	Radio	Yes	Yes / No	Whether the receiver's identity was checked
Handover Status	Handover_Status	Drop Down	Yes	Pending / Handed Over	Drives Returned Items and dashboard card
Handover Remarks	Handover_Remarks	Multi Line	No	—	Notes on the handover

Table 4.8 lists the proposed specification of the Item Handover form (link name Item_Handover). Link names, mandatory settings, validations and drop-down values are to be verified in Zoho Creator.

Table 4.9: Field Specification – Item Status Form

Field Label	Proposed Link Name	Field Type	Mandatory (design)	Validation / Default	Remarks
Status Log ID	Status_Log_ID	Auto Number	System	Auto-generated	Identifier of the status entry
Item Type	Item_Type	Radio	Yes	Lost / Found	Selects which item Lookup applies
Lost Item	Lost_Item	Lookup (Lost Items)	No	Used when Item Type is Lost	Relationship to Lost Items
Found Item	Found_Item	Lookup (Found Items)	No	Used when Item Type is Found	Relationship to Found Items
Current Status	Current_Status	Drop Down	Yes	Lost / Found / Possible Match / Under Claim / Returned / Closed	Status recorded at this time
Updated By	Updated_By	Lookup (Administrator Management)	Yes	Selects an existing administrator	Person who recorded the change
Status Date	Status_Date	Date-Time	No	Default: current datetime	When the status changed
Status Remarks	Status_Remarks	Multi Line	No	—	Reason for the change

Table 4.9 lists the proposed specification of the Item Status form (link name Item_Status). Link names, mandatory settings, validations and drop-down values are to be verified in Zoho Creator.

Table 4.10: Field Specification – Notifications Form

Field Label	Proposed Link Name	Field Type	Mandatory (design)	Validation / Default	Remarks
Notification ID	Notification_ID	Auto Number	System	Auto-generated	Identifier of the notification record
Recipient	Recipient	Lookup (User Registration)	Yes	Selects an existing user	User to whom the notification is addressed
Related Lost Item	Related_Lost_Item	Lookup (Lost Items)	No	Optional	Lost item concerned
Related Found Item	Related_Found_Item	Lookup (Found Items)	No	Optional	Found item concerned
Notification Type	Notification_Type	Drop Down	Yes	Item Reported / Possible Match / Claim Received / Claim Approved / Claim Rejected / Item Returned / Reminder	Kind of event
Field Label	**Proposed Link Name**	**Field Type**	**Mandatory (design)**	**Validation / Default**	**Remarks**
Message	Message	Multi Line	Yes	Not empty	Text shown to the recipient
Notification Date	Notification_Date	Date-Time	No	Default: current datetime (DS-10)	When the notification was created
Read Status	Read_Status	Drop Down	Yes	Default: Unread (DS-10); Unread / Read	Whether the user has seen it

Table 4.10 lists the proposed specification of the Notifications form (link name Notifications). Link names, mandatory settings, validations and drop-down values are to be verified in Zoho Creator.

Table 4.11: Field Specification – Administrator Management Form

Field Label	Proposed Link Name	Field Type	Mandatory (design)	Validation / Default	Remarks
Admin Name	Admin_Name	Single Line	Yes	Not empty	Name of the administrator
Admin Email	Admin_Email	Email	Yes	Valid email format	Used to identify the administrator
Designation	Designation	Single Line	Yes	Not empty	Post held in the college

Contact Number	Contact_Number	Phone	Yes	Valid phone format	Official contact number
Office Location	Office_Location	Single Line	No	—	Where the administrator can be met
Admin Status	Admin_Status	Drop Down	Yes	Default: Active; Active / Inactive	Whether the person is currently serving

Table 4.11 lists the proposed specification of the Administrator Management form (link name Administrator_Management). Link names, mandatory settings, validations and drop-down values are to be verified in Zoho Creator.

4.4 Process and Workflow Design

The portal supports one main process, the life cycle of a lost or found item, which is conceptually eight steps long: (1) a student or staff member reports a lost item; (2) another user reports a found item; (3) the administrator or authorised stakeholder reviews the available records; (4) the item is searched and identified; (5) a claim request is submitted; (6) the claim is verified; (7) the item is returned to its rightful owner; and
(8) the item status is updated. Table 4.12 classifies each step according to whether it is performed manually by a person, is a configured form behaviour, or is a proposed automation. The classification is an honest statement of the design, and nothing in the "proposed automation" column is claimed to be working.

Table 4.12: Classification of Process Steps

Step	Process Step	Manual / Form Entry	Configured Form Behaviour	Proposed Automation (not claimed)
1	Report lost item	User completes the Lost Items form	Auto Number; drop-down lists; Lookup to user	Default status and date; date check (DS-02, DS-03); Item Reported notification (DS-10)
2	Report found item	User completes the Found Items form	Auto Number; drop-down lists; Lookup to user	Default status and date; date check (DS-04, DS-05); notification record
3	Review records	Administrator opens reports and dashboard	Report listing and filters	—
4	Search and identify	Administrator or user searches and filters reports; administrator records the match in Item Tracking	Report search / filter panel	Automatic matching is a future enhancement (Section 9.2)

5	Submit claim	Claimant completes the Claim Requests form	Lookups to found item and claimant	Default Pending status; duplicate-claim check; found item set to Under Claim (DS-06, DS-07)
6	Verify claim	Administrator completes Claim Verification	Lookups to claim and administrator	Claim Status follows result; notification (DS-08)
7	Return item	Administrator completes Item Handover	Lookups to verification and administrator	Item statuses set to Returned; claim completed (DS-09)
8	Update status	Administrator may add an Item Status entry	Item Status form with Lookups	Status log entry created with DS-09; further automation is future work

Figures 4.3 to 4.8 show the process flows. Solid boxes represent manual or configured steps, and dashed boxes represent proposed automation that is to be verified in the application.

4.4.1 User Registration Process

A user opens the User Registration form and enters the name, register number or ID, user type, email, phone number, department and year. The form checks that the mandatory fields are complete. The proposed script DS-01 checks whether the registerdeployed number already exists and, if it does, rejects the entry. Otherwise the record is saved with the registration date and the user becomes available in the Lookup fields of the other forms.

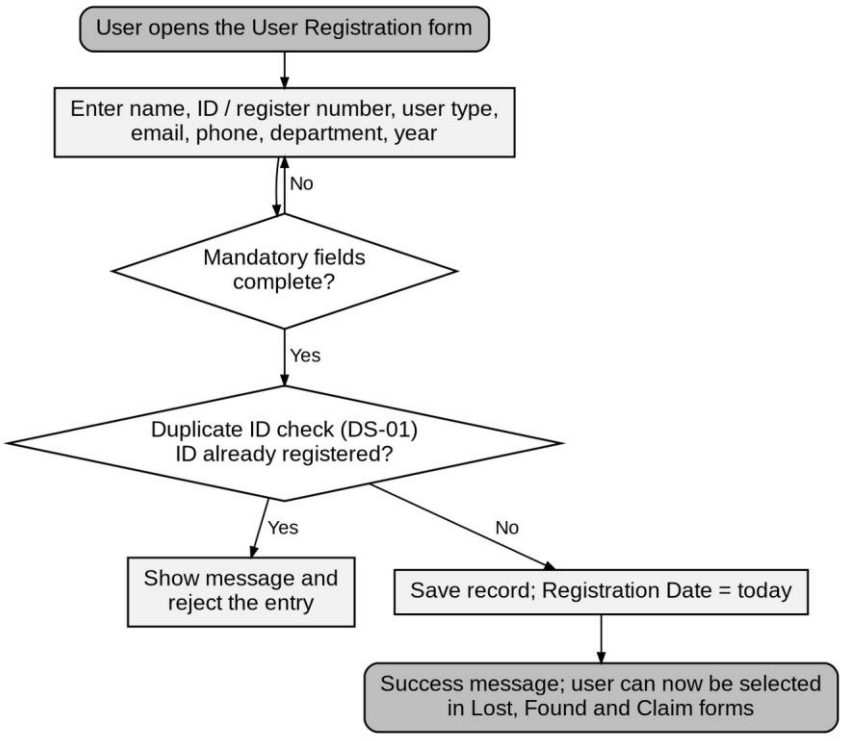


Figure 4.3: User Registration Process Flow

4.4.2 Lost Item Reporting Process

The reporting user selects himself or herself in the Reported By Lookup, chooses the category and enters the item name, description, lost location and lost date, and optionally uploads an image. If the entry is valid, the record is saved with a generated Lost Item ID and the default status Lost. A proposed notification record informs the persons concerned. The record then appears in the Lost Items Report and is counted in the Total Lost Items card.

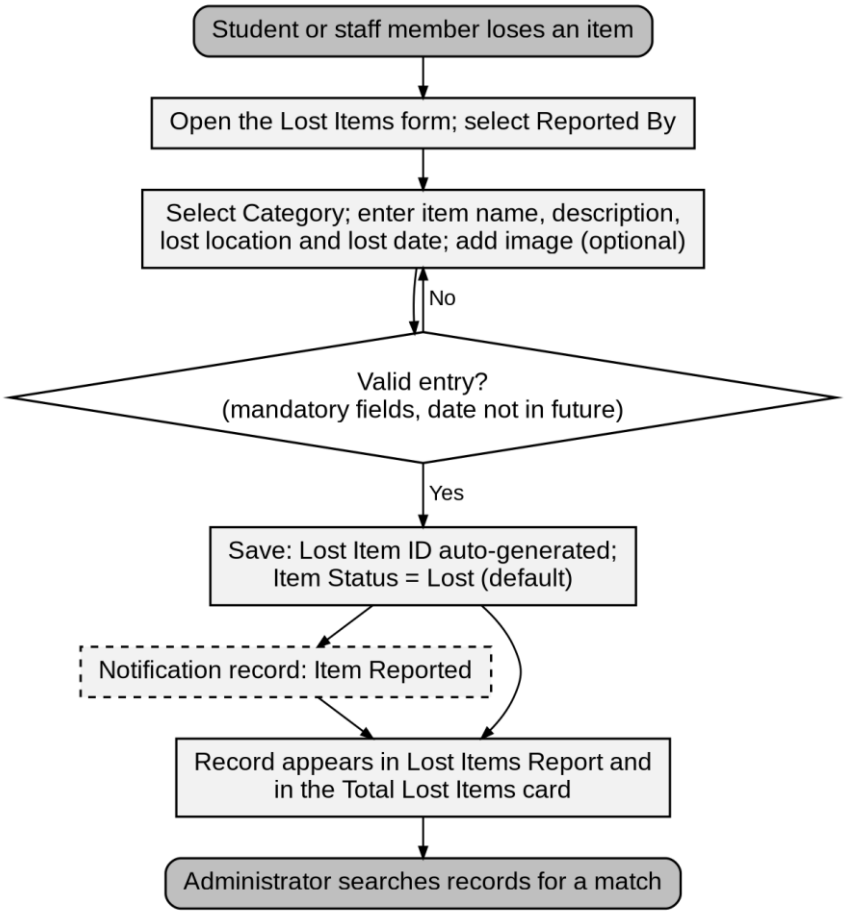


Figure 4.4: Lost Item Reporting Process Flow

4.4.3 Found Item Reporting Process

The finder reports the item in the same way, with the additional Storage Location that tells the owner where the item is held. After the record is saved, the administrator compares the item with the lost records using the search and filter facility. If a possible owner is identified, the administrator records the possible match in the Item Tracking form; otherwise the item stays in storage with status Found.

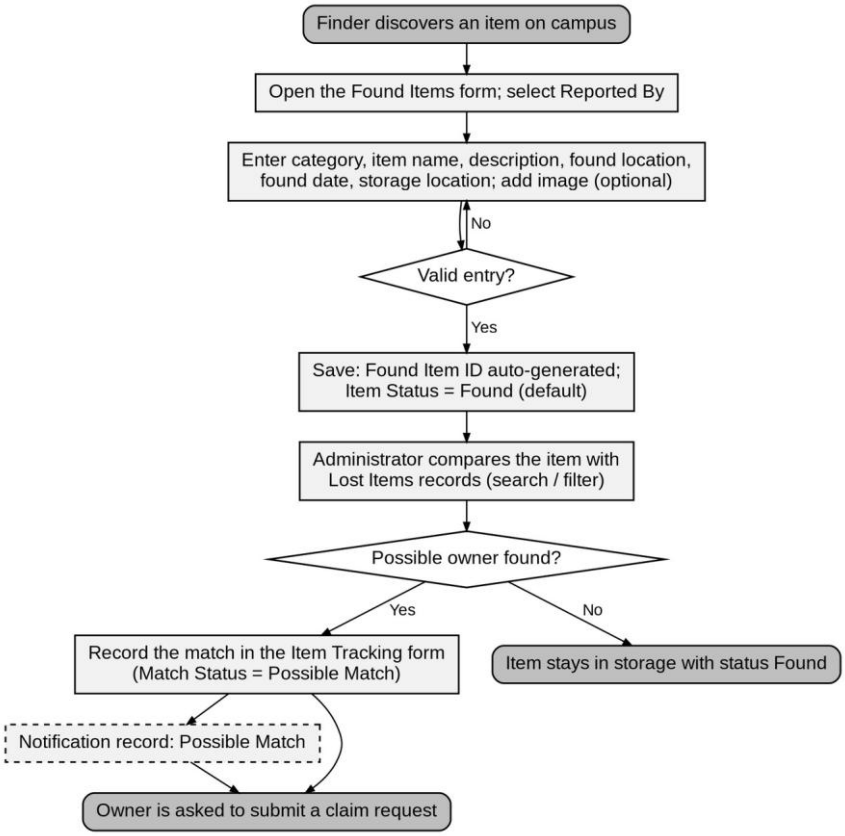


Figure 4.5: Found Item Reporting Process Flow

4.4.4 Claim Submission Process

An owner who finds the item in the Found Items Report opens the Claim Requests form, selects the found item and the claimant, and describes the item in a way that only the owner would know. The proposed scripts check that the description is present and thatdeployed the claimant has no other pending claim on the same item. The record is saved with the status Pending and appears in the Pending Claims report.

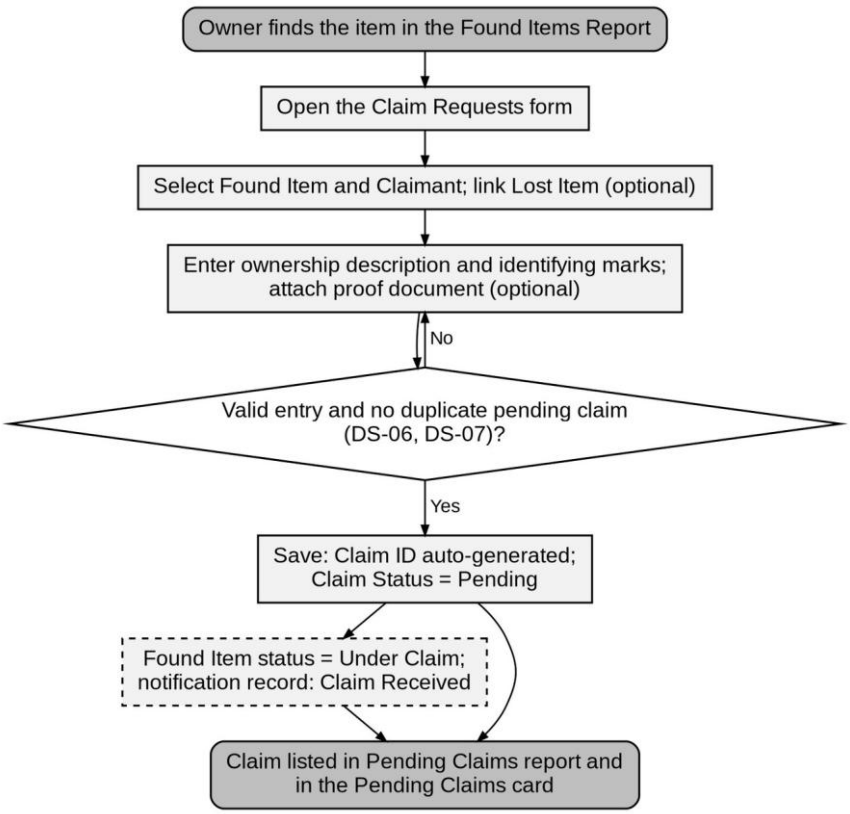


Figure 4.6: Claim Submission Process Flow

4.4.5 Claim Verification Process (Conceptual)

The administrator reviews the pending claim, compares the ownership description, identifying marks and any proof document with the found item, and records the outcome in the Claim Verification form. If ownership is established the result is Approved; otherwise it is Rejected, with remarks. Preventing false claims is the main purpose of this step. The automatic change of the claim status is a proposed script (DS-deployed 08); without it, the administrator changes the Claim Status manually.

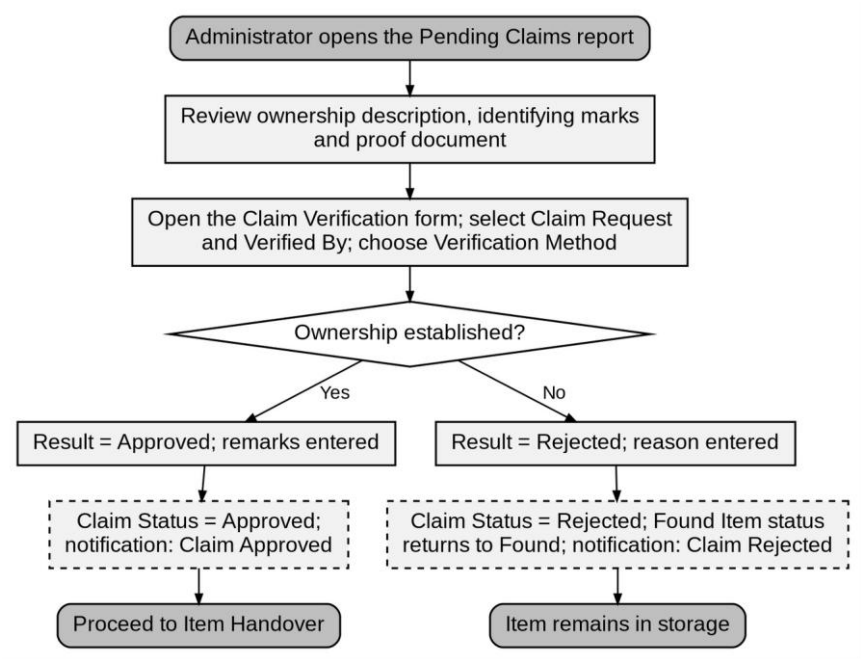


Figure 4.7: Claim Verification Conceptual Process Flow

4.4.6 Item Return Process

For an approved claim the administrator opens the Item Handover form, selects the verification record and the administrator who releases the item, enters the handover date and confirms that the receiver's identity was checked. When the record is saved, the proposed script DS-09 changes the item statuses to Returned and completes the claim;deployed the dashboard card Successfully Returned Items then counts the new record.

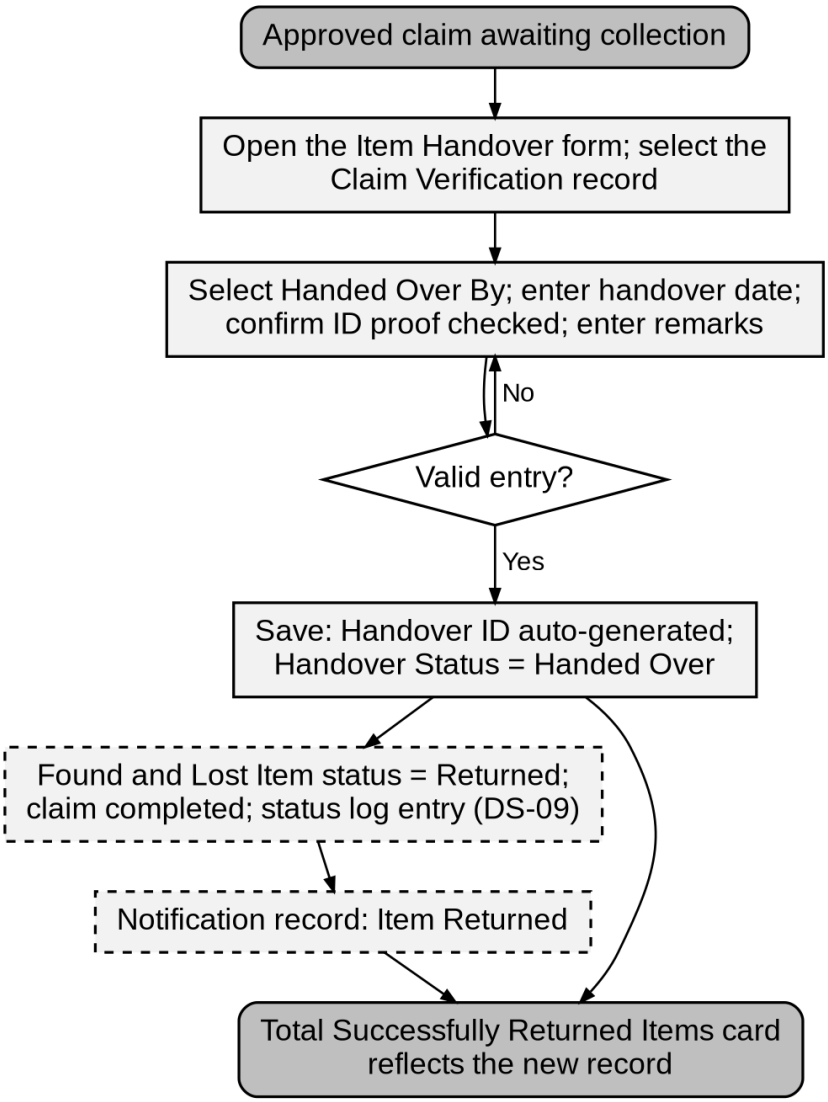


Figure 4.8: Item Return Process Flow

4.5 User Interface and Navigation Design

The navigation of the application is organised in sections that correspond to the functional areas and their reports, together with a dashboard section. Figure 4.9 shows the navigation structure. The sidebar of the dashboard is a vertical dark navy-blue panel with a professional college-administration appearance, and it lists the sections: Dashboard, Users, Lost Items, Found Items, Item Tracking, Claims, Handover, Item Status, Notifications and Administration. Each section leads only to forms and reports that exist in the application, and the menu items visible to a user depend on the role.

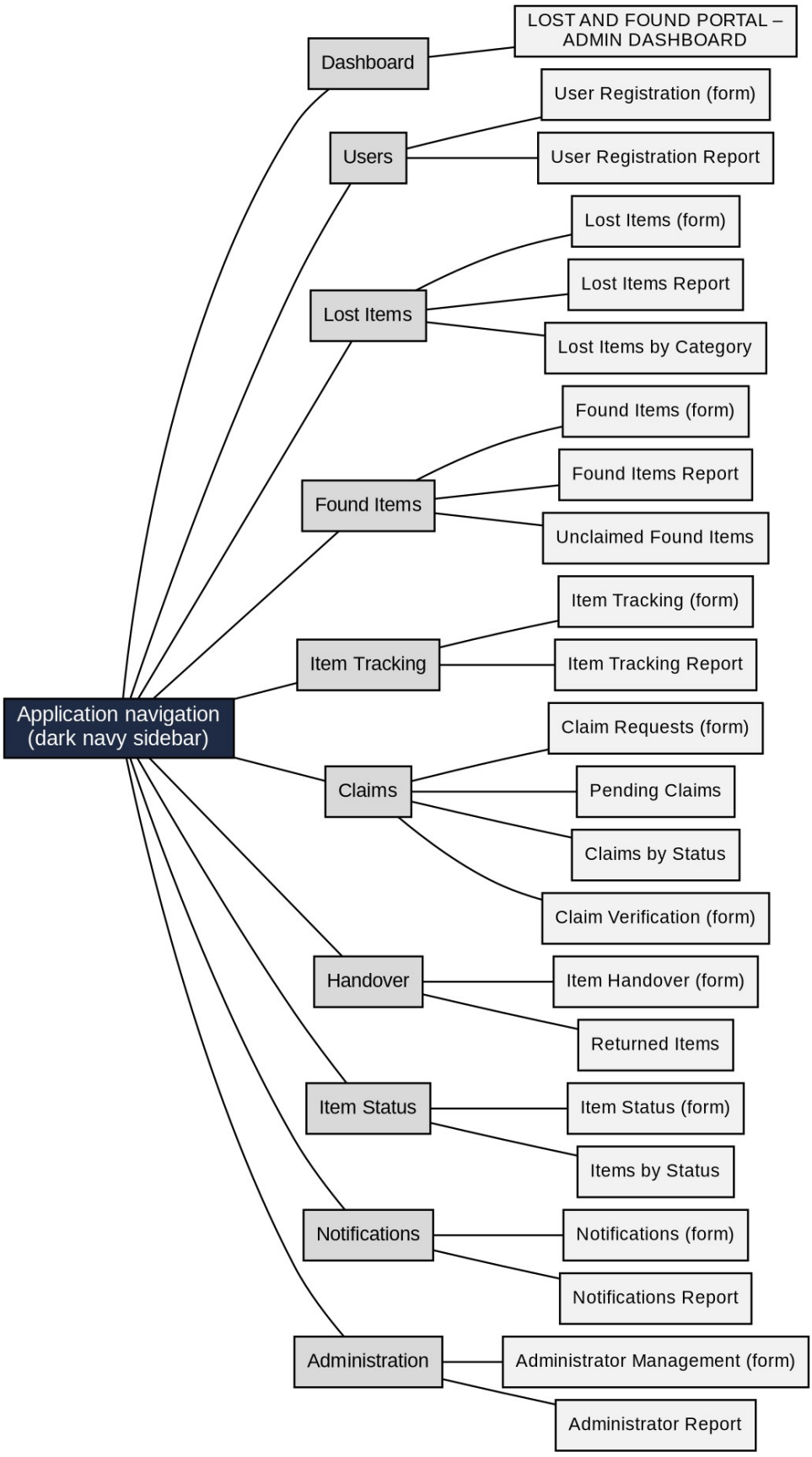

Application navigation (dark navy sidebar)
Dashboard
LOST AND FOUND PORTAL – ADMIN DASHBOARD
Users
User Registration (form)
User Registration Report
Lost Items
Lost Items (form)
Lost Items Report
Lost Items by Category
Found Items
Found Items (form)
Found Items Report
Unclaimed Found Items
Item Tracking
Item Tracking (form)
Item Tracking Report
Claims
Claim Requests (form)
Pending Claims
Claims by Status
Claim Verification (form)
Handover
Item Handover (form)
Returned Items
Item Status
Item Status (form)
Items by Status
Notifications
Notifications (form)
Notifications Report
Administration
Administrator Management (form)
Administrator Report

Figure 4.9: Application Navigation Structure The wireframe of the LOST AND FOUND PORTAL – ADMIN DASHBOARD is described from top to bottom and is shown in Figure 4.10. The left side holds the dark navy-blue sidebar. The header carries the dashboard name. Below it is a row of four KPI cards: Total Lost Items, Total Found Items, Pending Claims and Successfully Returned Items. The middle section holds four horizontal bar charts: Lost and Found Item Overview, Items by Category, Item Status Distribution and Claim Request Overview. The bottom section holds the lists Recently Reported Lost Items, Recently Reported Found Items and Recent Claim Requests, and a Quick Links panel whose buttons open the forms for reporting a lost item, reporting a found item, submitting a claim, verifying a claim and recording a handover. The wireframe is a layout sketch and contains no real data; the bars shown are schematic.

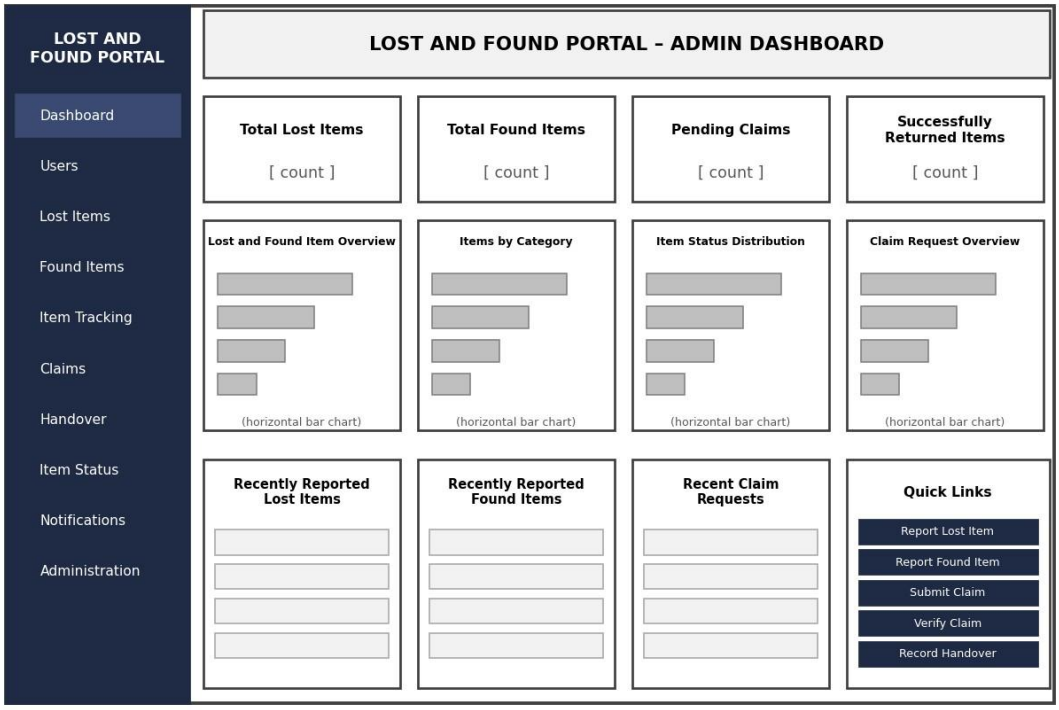


Figure 4.10: Wireframe of the LOST AND FOUND PORTAL – ADMIN DASHBOARD

4.6 Access Control Design

Table 4.13 defines the permission matrix that Chapter 5 is intended to implement, using A = Add, E = Edit, V = View, D = Delete and — = No access. The phrase “own” means that a record-level criterion restricts the user to records that the user added or that relate to the user. Students and faculty / staff are shown together because their needs are the same in the present design. The matrix is a conceptual design; the

permissions that have actually been configured and tested are reported separately in Section 5.8, and no role or permission profile is claimed to exist on the basis of this table.

Table 4.13: Role–Permission Matrix (Conceptual Design)

Component	Administrator	Student / Faculty / Staff
Form: User Registration	A E V D	A V (own)
Form: Administrator Management	A E V D	—
Form: Lost Items	A E V D	A V (own)
Form: Found Items	A E V D	A; V (found items without finder contact details)
Form: Item Tracking	A E V D	—
Form: Claim Requests	A E V D	A V (own)
Form: Claim Verification	A E V D	V (own claim; result and remarks)
Form: Item Handover	A E V D	V (own handover)
Form: Item Status	A E V D	V (own items)
Form: Notifications	A E V D	V (own; Read Status only)
Reports: all form reports	V	V (own, where permitted)
Reports: Pending Claims, Claims by Status, Returned Items, Items by Status	V	—
Page: LOST AND FOUND PORTAL – ADMIN DASHBOARD	V	—

The access design is concerned with three kinds of information. First, student details such as name, register number, email and phone number must be visible only to the administrator and to the person concerned. Second, contact information must not be disclosed to a person who has merely seen an item in a report; the owner and the finder should be put in touch through the administrator. Third, claim information – the ownership description, identifying marks and proof documents – must be visible only to the claimant and the administrator, because the description is the evidence by which a false claim is detected. Allowing an unverified person to read the full description of a found item would make it easy to make a false claim, and the Found Items report shown to ordinary users should therefore carry only the general description. Whether these restrictions have been configured in the application is reported in Section 5.8.

CHAPTER 5

IMPLEMENTATION

This chapter describes the implementation of the Lost and Found Portal on Zoho Creator. The ten forms, the sixteen reports and the administrative dashboard are described as specified in the project details. The Deluge scripts, workflows, notification records and role configuration are described as an implementation description derived from the design; their actual configuration and testing status must be confirmed against the live application and recorded in Appendix A. Each figure has a text description of the corresponding screen and a reserved space in which the student should paste the actual screenshot.

5.1 Application Setup

The application **Lost and Found Portal** was created in a Zoho Creator workspace as a new application. Zoho Creator is the low-code platform of Zoho Corporation on which forms, reports, pages, workflows and scripts are built in a browser-based builder, and it hosts the finished application [2]. The application was created with the assistance of Zia.

Zia was used as an AI-assisted application builder: a description of the campus lost-andfound process was given to the assistant, which proposed a first set of forms and fields. The proposals were reviewed by the developer, and the final set of ten forms in this report is the one specified for the project. This report does not assume that every suggestion made by Zia was accepted or implemented; the forms, fields and Lookup relationships that were created, renamed, corrected or rejected after review are to be listed here from the development log. Zia assists the creation of the application only; it is not a run-time feature of the application (Section 5.9).

The application sections are organised to follow the navigation described in Section 4.5, so that every form has its entry form and its related reports under one section. The theme uses a dark navy-blue sidebar for the dashboard and a plain light content area for readability. On opening the application, the user lands on the menu on the left side of the screen, which lists the Dashboard and the sections for Users, Lost Items, Found Items, Item Tracking, Claims, Handover, Item Status, Notifications and Administration. The sections shown depend on the permission profile of the logged-in user.

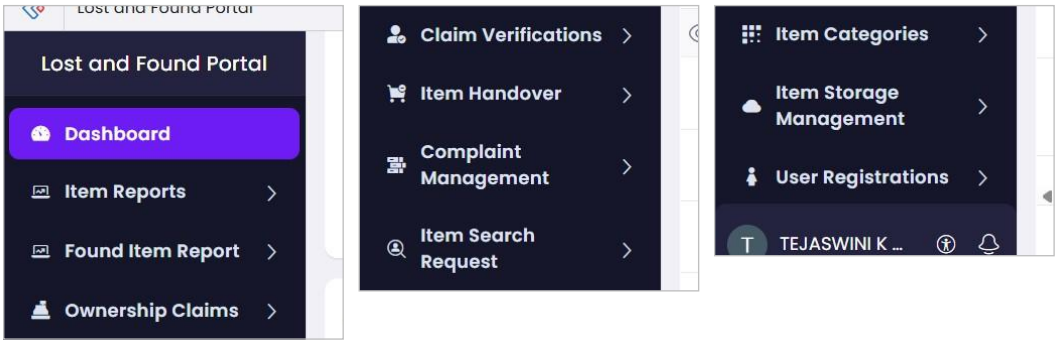


Figure 5.1: Application Home and Navigation Sections

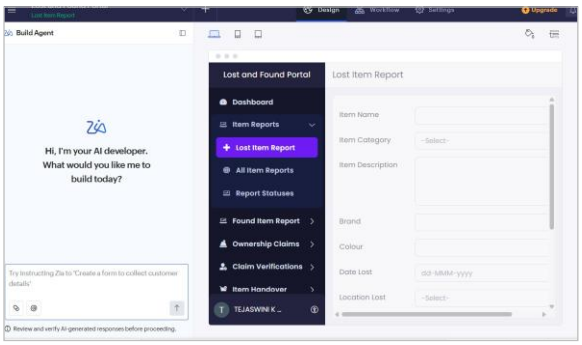


Figure 5.2: Zia-Assisted Application Creation Screen

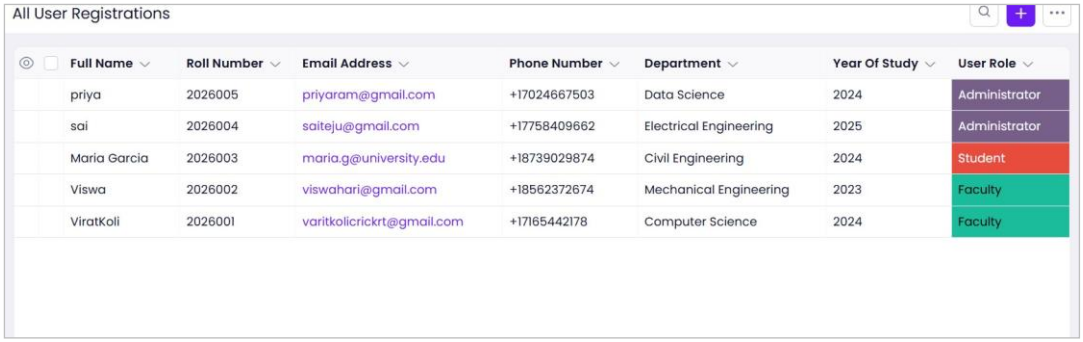
5.2 Forms and Data Capture

The ten forms implement the data model of Chapter 4. Each form is described below with its purpose, fields and field types, Lookup relationships, data captured and role in the portal. The field properties follow Tables 4.2 to 4.11. Mandatory fields are marked as required in the form builder, and the Lookup fields display the user name or item name for easy selection. Field-level validations are provided either by field properties (for example, the no-duplicate-values option for the Register Number / ID) or by the Deluge scripts of Section 5.5. The forms use a single-column layout to suit mobile screens, and no conditional layout rules were specified in the project brief. No form beyond these ten is claimed.

5.2.1 User Registration Form

Purpose. The form records every student, faculty member and staff member who uses the portal and is the master source of person details. **Fields and types.** It has eight fields: User Name (Single Line), Register Number / ID (Single Line, unique), User Type (Drop Down), Email (Email), Phone Number (Phone), Department (Drop Down), Year of Study (Drop Down) and Registration Date (Date). **Lookup relationships.** It has no Lookup field; it is the parent of the Lookups in Lost Items, Found Items, Claim

Requests and Notifications. **Data captured.** Identity and contact details of each user, entered once. **Role in the portal.** Allows other forms to refer to a user without repeating personal information.

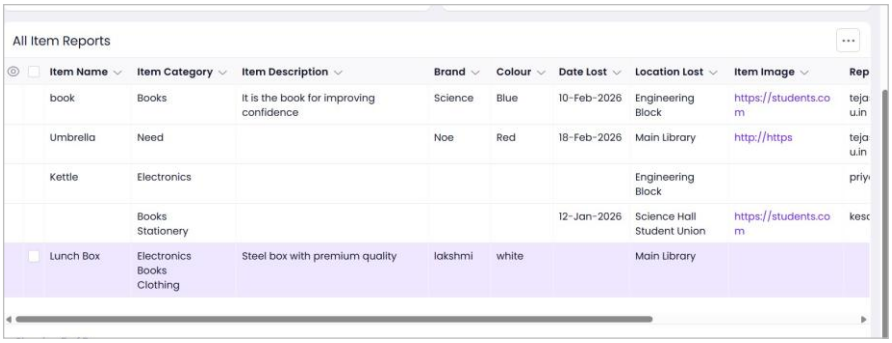


Full Name	Roll Number	Email Address	Phone Number	Department	Year Of Study	User Role
priya	2026005	priyaram@gmail.com	+17024667503	Data Science	2024	Administrator
sai	2026004	saiteju@gmail.com	+17758409662	Electrical Engineering	2025	Administrator
Maria Garcia	2026003	maria.g@university.edu	+18739029874	Civil Engineering	2024	Student
Viswa	2026002	viswahari@gmail.com	+18562372674	Mechanical Engineering	2023	Faculty
ViratKoli	2026001	varitkolicrickrt@gmail.com	+17165442178	Computer Science	2024	Faculty

Figure 5.3: User Registration Form

5.2.2 Lost Items Form

Purpose. The form records every report of a lost belonging. **Fields and types.** It has ten fields: Lost Item ID (Auto Number), Reported By (Lookup), Item Name (Single Line), Category (Drop Down), Item Description (Multi Line), Lost Location (Single Line), Lost Date (Date), Item Image (Image, optional), Report Date (Date) and Item Status (Drop Down). **Lookup relationships.** Reported By refers to User Registration. **Data captured.** What was lost, its category and description, where and when it was lost, an optional image, and the current status. **Role in the portal.** Provides the record against which found items are compared and the count shown in the Total Lost Items card.



Item Name	Item Category	Item Description	Brand	Colour	Date Lost	Location Lost	Item Image	Rep
book	Books	It is the book for improving confidence	Science	Blue	10-Feb-2026	Engineering Block	https://students.com	teja u.in
Umbrella	Need		Noe	Red	18-Feb-2026	Main Library	http://https	teja u.in
Kettle	Electronics					Engineering Block		priy
	Books Stationery				12-Jan-2026	Science Hall Student Union	https://students.com	kesc
Lunch Box	Electronics Books Clothing	Steel box with premium quality	lakshmi	white		Main Library		

Figure 5.4: Lost Items Form

5.2.3 Found Items Form

Purpose. The form records every report of a found belonging and the place where it is kept. **Fields and types.** It has eleven fields: Found Item ID (Auto Number), Reported By (Lookup), Item Name, Category, Item Description, Found Location, Found Date, Storage Location, Item Image (Image, optional), Report Date and Item Status. **Lookup relationships.** Reported By refers to User Registration. **Data captured.** What was found, where and when, where it is stored, an optional image and the current status.

Role in the portal. Provides the inventory of items held for owners and the count shown in the Total Found Items card.

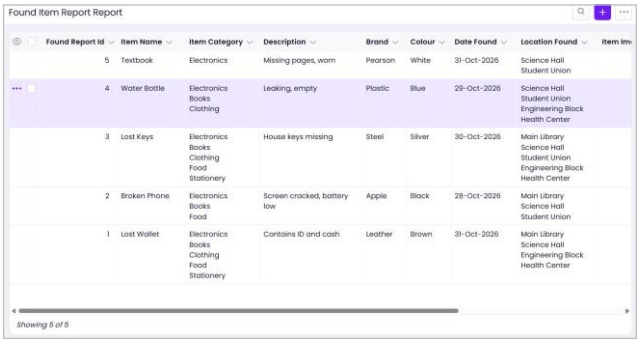


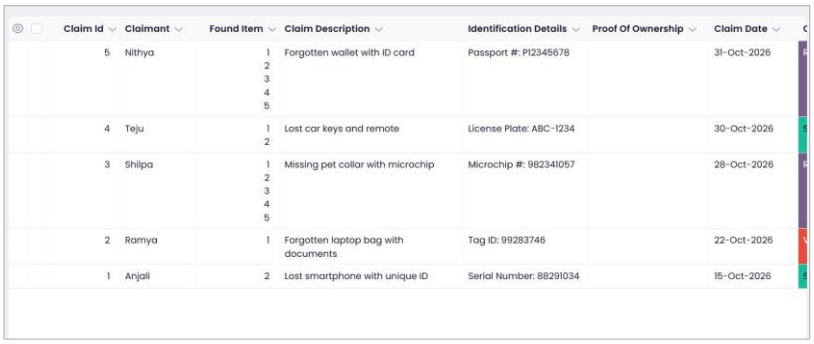
Figure 5.5: Found Items Form

5.2.4 Item Tracking Form

Purpose. The form supports the search-and-tracking function by recording the match that the administrator identifies between a lost item and a found item. **Fields and types.** It has seven fields: Tracking ID (Auto Number), Lost Item (Lookup), Found Item (Lookup), Tracked By (Lookup), Tracking Date (Date-Time), Match Status (Drop Down) and Match Remarks (Multi Line). **Lookup relationships.** Lost Item refers to Lost Items, Found Item to Found Items and Tracked By to Administrator Management. **Data captured.** The two items judged to correspond, the administrator, the date and the status and basis of the match. **Role in the portal.** Connects a lost report with a found report without changing either record, and shows who made the connection.

5.2.5 Claim Requests Form

Purpose. The form records a request by a user to receive a found item. **Fields and types.** It has nine fields: Claim ID (Auto Number), Found Item (Lookup), Claimant (Lookup), Lost Item (Lookup, optional), Ownership Description (Multi Line), Identifying Marks (Multi Line), Proof Document (File Upload, optional), Claim Date (Date) and Claim Status (Drop Down). **Lookup relationships.** Found Item refers to Found Items, Claimant to User Registration and Lost Item to Lost Items. **Data captured.** The claimant's evidence of ownership and the progress of the claim. **Role in the portal.** Starts the claim cycle and feeds the Pending Claims report and card.

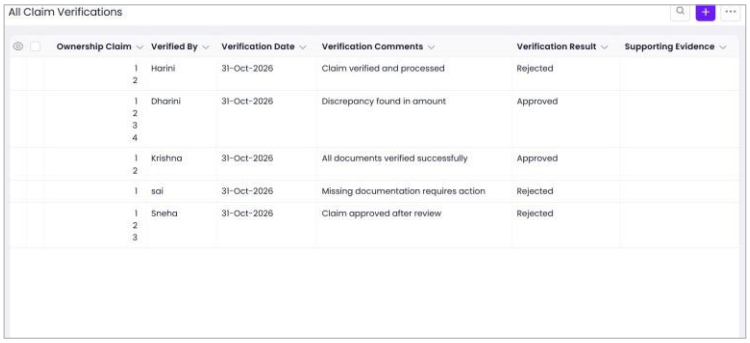


Claim Id	Claimant	Found Item	Claim Description	Identification Details	Proof Of Ownership	Claim Date
5	Nithya	1 2 3 4 5	Forgotten wallet with ID card	Passport #: P12345678		31-Oct-2026
4	Teju	1 2	Lost car keys and remote	License Plate: ABC-1234		30-Oct-2026
3	Shilpa	1 2 3 4 5	Missing pet collar with microchip	Microchip #: 982341057		28-Oct-2026
2	Ramya	1	Forgotten laptop bag with documents	Tag ID: 99283746		22-Oct-2026
1	Anjali	2	Lost smartphone with unique ID	Serial Number: 88291034		15-Oct-2026

Figure 5.7: Claim Requests Form

5.2.6 Claim Verification Form

Purpose. The form records how and by whom a claim was checked and the result. **Fields and types.** It has seven fields: Verification ID (Auto Number), Claim Request (Lookup), Verified By (Lookup), Verification Date (Date), Verification Method (Drop Down), Verification Result (Drop Down) and Verification Remarks (Multi Line). **Lookup relationships.** Claim Request refers to Claim Requests and Verified By to Administrator Management. **Data captured.** The method, result, date, verifier and remarks of each verification. **Role in the portal.** Prevents false claims by making every approval or rejection a recorded, attributable decision.



All Claim Verifications

Ownership Claim	Verified By	Verification Date	Verification Comments	Verification Result	Supporting Evidence
1 2	Harini	31-Oct-2026	Claim verified and processed	Rejected	
1 2 3 4	Dharini	31-Oct-2026	Discrepancy found in amount	Approved	
1 2	Krishna	31-Oct-2026	All documents verified successfully	Approved	
1	sai	31-Oct-2026	Missing documentation requires action	Rejected	
1 2 3	Sneha	31-Oct-2026	Claim approved after review	Rejected	

Figure 5.8: Claim Verification Form

5.2.7 Item Handover Form

Purpose. The form records the release of an item to its verified owner. **Fields and types.** It has seven fields: Handover ID (Auto Number), Claim Verification (Lookup), Handed Over By (Lookup), Handover Date (Date-Time), ID Proof Checked (Radio), Handover Status (Drop Down) and Handover Remarks (Multi Line). **Lookup relationships.** Claim Verification refers to Claim Verification and Handed Over By to Administrator Management. **Data captured.** When, by whom and after what identity check the item was released. **Role in the portal.** Closes the claim and supplies the Successfully Returned Items card.

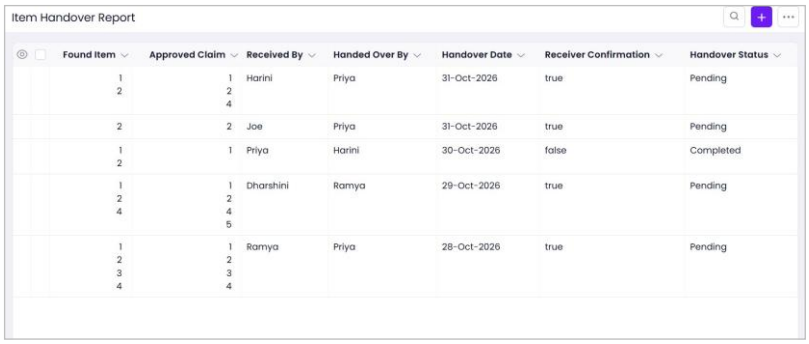


Figure 5.9: Item Handover Form

5.2.8 Item Status Form

Purpose. The form keeps a dated log of the status of lost and found items. **Fields and types.** It has eight fields: Status Log ID (Auto Number), Item Type (Radio), Lost Item (Lookup), Found Item (Lookup), Current Status (Drop Down), Updated By (Lookup), Status Date (Date-Time) and Status Remarks (Multi Line). **Lookup relationships.** Lost Item refers to Lost Items, Found Item to Found Items and Updated By to Administrator Management; only the Lookup that matches the Item Type is used. **Data captured.** The status, the time, the person and the reason for each change. **Role in the portal.** Gives the history behind the current Item Status of an item and feeds the Item Status Distribution chart.

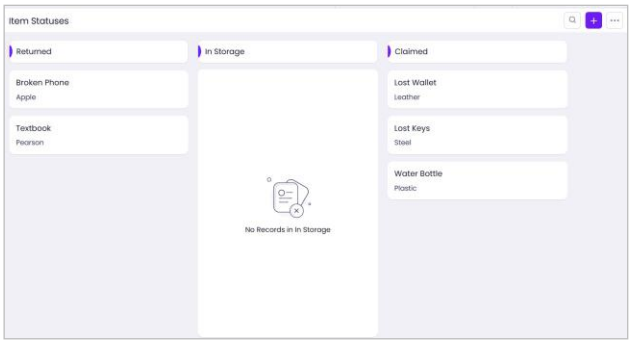
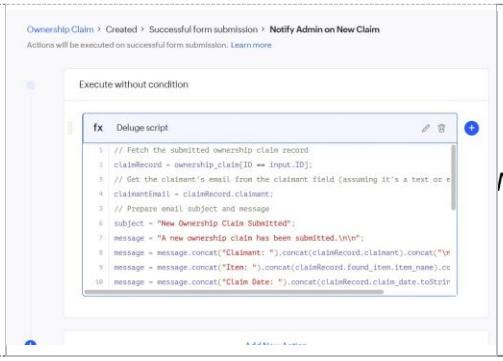


Figure 5.10: Item Status Form

5.2.9 Notifications Form

Purpose. The form stores a message addressed to a user about an event concerning an item. **Fields and types.** It has eight fields: Notification ID (Auto Number), Recipient (Lookup), Related Lost Item (Lookup), Related Found Item (Lookup), Notification Type (Drop Down), Message (Multi Line), Notification Date (Date-Time) and Read Status (Drop Down). **Lookup relationships.** Recipient refers to User Registration; the related item Lookups refer to Lost Items and Found Items. **Data captured.** The type, text, time and read state of each notification. **Role in the portal.** Provides in-application

notifications; delivery by email is not part of this form and is proposed separately (Section 5.7).

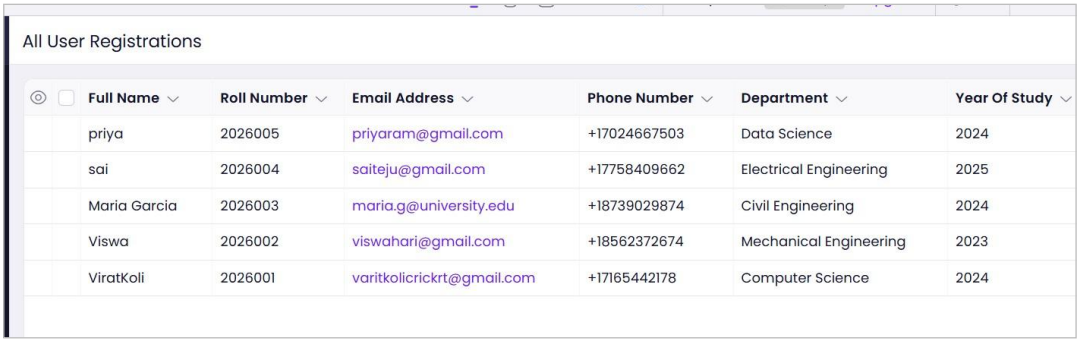


[INSERT ACTUA *NS FORM HERE]*

Figure 5.11: Notifications Form

5.2.10 Administrator Management Form

Purpose. The form records the persons who administer the portal. **Fields and types.** It has six fields: Admin Name (Single Line), Admin Email (Email), Designation (Single Line), Contact Number (Phone), Office Location (Single Line) and Admin Status (Drop Down). **Lookup relationships.** It has no Lookup field; it is the parent of the Lookups Tracked By, Verified By, Handed Over By and Updated By. **Data captured.** Identity and office details of each administrator. **Role in the portal.** Makes every administrative action traceable to a named person.


All User Registrations

Full Name	Roll Number	Email Address	Phone Number	Department	Year Of Study
priya	2026005	priyaram@gmail.com	+17024667503	Data Science	2024
sai	2026004	saiteju@gmail.com	+17758409662	Electrical Engineering	2025
Maria Garcia	2026003	maria.g@university.edu	+18739029874	Civil Engineering	2024
Viswa	2026002	viswahari@gmail.com	+18562372674	Mechanical Engineering	2023
ViratKoli	2026001	varitkolicrickrt@gmail.com	+17165442178	Computer Science	2024

Figure 5.12: Administrator Management Form

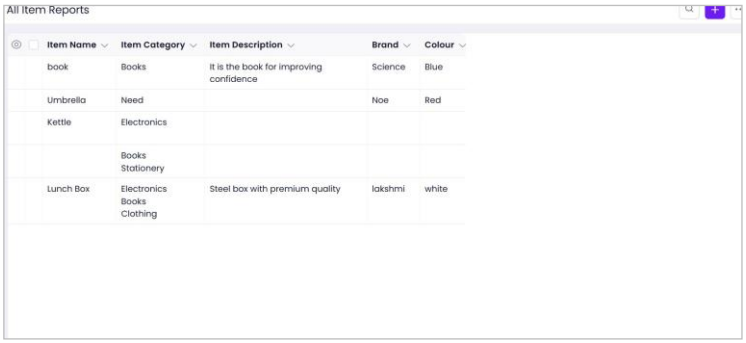
5.3 Reports

Sixteen reports are provided: ten default reports (one for each form) and six reports that apply criteria or grouping. Table 5.1 lists them. The report types, criteria and groupings are the implementation description; the exact report type of each grouped report (list report with grouping, chart or pivot) must be confirmed in the application. Reports are used by the roles shown, subject to the record-level restrictions of Section 4.6. **Table 5.1: Reports Implemented**

Report Name	Type	Source Form	Filter / Grouping / Sorting	Roles

User Registration Report	List	User Registration	Sort by Registration Date descending	Administrator; User (own)
Lost Items Report	List	Lost Items	Sort by Report Date descending; search and filter by Category, Lost Location, Lost Date, Item Status	Administrator; User (own)
Found Items Report	List	Found Items	Sort by Report Date descending; search and filter by Category, Found Location, Found Date, Item Status	Administrator; Users (general fields only)
Item Tracking Report	List	Item Tracking	Sort by Tracking Date descending	Administrator
Claim Requests Report	List	Claim Requests	Sort by Claim Date descending	Administrator; Claimant (own)
Claim Verification Report	List	Claim Verification	Sort by Verification Date descending	Administrator; Claimant (own)
Item Handover Report	List	Item Handover	Sort by Handover Date descending	Administrator; User (own)
Item Status Report	List	Item Status	Sort by Status Date descending	Administrator; User (own)
Notifications Report	List	Notifications	Sort by Notification Date descending	Administrator; Recipient (own)
Administrator Management Report	List	Administrator Management	Sort by Admin Name	Administrator
Pending Claims	List with criteria	Claim Requests	Criteria: Claim Status is Pending	Administrator
Unclaimed Found Items	List with criteria	Found Items	Criteria: Item Status is Found	Administrator; Users (as permitted)
Returned Items	List with criteria	Item Handover	Criteria: Handover Status is Handed Over	Administrator
Lost Items by Category	Chart / grouped list	Lost Items	Group by Category; count of records	Administrator
Items by Status	Chart / grouped list	Found Items (and Lost Items)	Group by Item Status; count of records	Administrator
Claims by Status	Chart / grouped list	Claim Requests	Group by Claim Status; count of records	Administrator

Item search is provided through the search and filter facility of the Lost Items Report and the Found Items Report, which lets a user look for an item by keyword or by Category, location, date and status. This is a facility of the report component of Zoho Creator and not a custom search page. The Pending Claims report is a list report on the Claim Requests form with the criterion Claim Status equal to Pending. Its columns are Claim ID, Found Item, Claimant, Claim Date and Claim Status. The administrator opens a record from this report to read the ownership description and then proceeds to the Claim Verification form.



5.4 Pages and Dashboards

The LOST AND FOUND PORTAL – ADMIN DASHBOARD is a page built with the page builder of Zoho Creator and is intended for the Administrator. It combines a sidebar, KPI cards, charts, embedded lists and quick links. The components and the calculation behind each are described below. No numerical value is stated here, because the values depend on the data entered and must be read from the live application; where no record exists a card shows zero. The build status of each component in the application is to be confirmed.

Sidebar. A dark navy-blue vertical sidebar with a professional college-administration appearance carries the section names Dashboard, Users, Lost Items, Found Items, Item Tracking, Claims, Handover, Item Status, Notifications and Administration. Each entry navigates to an existing form or report.

KPI cards. The four cards are calculated from actual records:

- **Total Lost Items** – the count of records in the Lost Items form.
- **Total Found Items** – the count of records in the Found Items form.
- **Pending Claims** – the count of records in the Claim Requests form whose Claim Status is Pending.
- **Successfully Returned Items** – the count of records in the Item Handover form whose Handover Status is Handed Over.

Middle section. Four horizontal bar charts are described: **Lost and Found Item Overview** compares the number of lost records and found records; **Items by Category** shows the number of items in each Category; **Item Status Distribution** shows the number of items in each Item Status; and **Claim Request Overview** shows the number of claims in each Claim Status.

Bottom section. The lists **Recently Reported Lost Items**, **Recently Reported Found Items** and **Recent Claim Requests** show the latest records sorted by date, and the **Quick Links** panel opens the forms for reporting a lost item, reporting a found item, submitting a claim request, recording a verification and recording a handover. Quick Links navigate only to forms and pages that exist in the application. Because the dashboard is a view over the same records, it is expected to update as new records are added; this dynamic behaviour is not claimed until it is verified (test cases TC-21 to TC23).

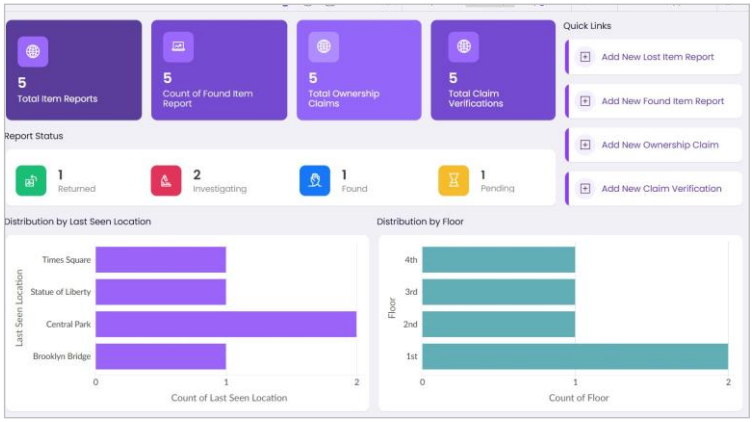


Figure 5.15: LOST AND FOUND PORTAL – ADMIN DASHBOARD

5.5 Deluge Scripting

Deluge is the scripting language of Zoho Creator, with statements for fetching, inserting and updating records, conditions, alerts and email [4]. The scripts implemented for the Lost and Found Portal are listed in Table 5.2. They use the link names of Chapter 4, were configured in the Zoho Creator application and have been verified there. Each Script ID of Table 5.2 is confirmed in Appendix A, and the complete listings appear in Appendix C.

Table 5.2: Deluge Scripts (Implemented and Verified)

Script ID	Location / Trigger	Purpose	Deluge Concepts Used

DS-01	User Registration → On Validate	Prevent a duplicate Register Number / ID (FR-02)	criteria fetch with count, if, alert, cancel submit
DS-02	Lost Items → On Load	Set default Item Status to Lost and Report Date to today (FR-04)	field assignment, zoho.currentdate
DS-03	Lost Items → On Validate	Reject a Lost Date later than today (FR-04)	date comparison, alert, cancel submit
DS-04	Found Items → On Load	Set default Item Status to Found and Report Date to today (FR-06)	field assignment, zoho.currentdate
DS-05	Found Items → On Validate	Reject a Found Date later than today (FR06)	date comparison, alert, cancel submit
DS-06	Claim Requests → On Load and On Validate	Set default status Pending and date; require an Ownership Description; refuse a duplicate pending claim (FR-11)	field assignment, string checks, criteria fetch with count
DS-07	Claim Requests → On Success	Set the found item to Under Claim and create a Claim Received notification record (FR-11, FR-15)	fetch, record update, insert into
DS-08	Claim Verification → On Success	Change the Claim Status according to the Verification Result and create a notification record (FR-12)	fetch, if–else, record update, insert into
DS-09	Item Handover → On Success	Set found and lost item to Returned, complete the claim and create an Item Returned notification (FR-13)	fetch, record update, insert into
DS-10	Lost Items and Found Items → On Success	Create an Item Reported notification record for the reporting user (FR-15)	insert into, zoho.currenttime
DS-11	Item Status → On Success	Apply the Current Status to the related lost or found item (FR-14)	fetch, if–else, record update

Code 5.1: Validation script DS-01 – duplicate Register Number / ID (verified in the application)

```
// DS-01 | Form: User_Registration | Event: On Validate
// Purpose: Block registration if the Register Number / ID already
exists
dupCount = User_Registration[Register_Number_ID ==
input.Register_Number_ID].count();
if(dupCount > 0)
{
    alert "This Register Number / ID is already registered.";
    cancel submit;
}
```

Code 5.2: Status and notification script DS-07 – claim received (verified in the application)

```
// DS-07 | Form: Claim_Requests | Event: On Success
// Purpose: Mark the found item as Under Claim and record a
```

```
notification
fRec = Found_Items[ID == input.Found_Item];
if(fRec.ID != null)
{
    fRec.Item_Status = "Under Claim";     insert into Notifications
    [
        Added_User = zoho.loginuser
        Recipient = input.Claimant
        Related_Found_Item = input.Found_Item
        Notification_Type = "Claim Received"
        Message = "Your claim request has been received and is
pending verification."
        Notification_Date = zoho.currenttime
        Read_Status = "Unread"
    ]
}
```

DS-01 counts the records of User Registration that have the entered Register Number / ID; if the count is above zero, an alert is shown and the submission is cancelled. The unique-value property of the field can achieve the same result without code, and the script is the low-code alternative that gives a customised message. DS-07 fetches the found item by its record identifier, changes its status and inserts a notification record for the claimant. No execution result is claimed here. The remaining scripts follow the same patterns and are given in Appendix C.

5.6 Workflows and Automation

In Zoho Creator, a form workflow runs scripts on form events, while schedule, approval, Blueprint and batch workflows serve other purposes. The Lost and Found Portal is a simple status-driven application, so form workflows are the principal mechanism. Table 5.3 lists the workflows intended for the ten forms, with the type, trigger, condition, actions and script references. Their configuration status must be confirmed in the application. If the workflows were not configured, the automated validation, status updating and notification processes described here were not implemented, and the corresponding steps are performed manually by the administrator as described in Section 4.4. Approval, Blueprint, schedule and batch workflows are not part of the project brief and are recorded in the table only to state that they are not planned.

Table 5.3: Workflows (Proposed; Configuration Status to be Verified)

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
User Validation	Form Workflow	On Validate (Add)	Register Number / ID exists	Block submission with alert	DS-01

Lost Item Defaults	Form Workflow	On Load	New record	Set Item Status = Lost; Report Date = today	DS-02
Lost Date Validation	Form Workflow	On Validate	Lost Date later than today	Block submission with alert	DS-03
Found Item Defaults	Form Workflow	On Load	New record	Set Item Status = Found; Report Date = today	DS-04
Found Date Validation	Form Workflow	On Validate	Found Date later than today	Block submission with alert	DS-05
Claim Defaults and Validation	Form Workflow	On Load; On Validate	New record; description empty; duplicate pending claim	Set Status = Pending; block if invalid	DS-06
Claim Submission Update	Form Workflow	On Success (Add)	Found item exists	Set found item to Under Claim; insert notification	DS-07
Verification Result Update	Form Workflow	On Success (Add)	Result is Approved or Rejected	Update Claim Status; insert notification	DS-08
Handover Update	Form Workflow	On Success (Add)	Handover Status = Handed Over	Set items to Returned; claim Completed; insert notification	DS-09
Item Report Notification	Form Workflow	On Success (Add) on Lost Items and Found Items	Reporting user exists	Insert Item Reported notification	DS-10
Status Synchronisation	Form Workflow	On Success (Add)	Item Type and Lookup present	Update status of the related item	DS-11
Approval / Blueprint / Schedule / Batch	Not planned	—	—	Not used in the ten-form design; see Section 9.2 for possible scheduled reminders for unclaimed items	—

The workflow configuration screen of the Claim Requests form lists the workflows by form event. Under On Load, the default-value script is attached; under On Validate, the description and duplicate checks are attached; and under On Success, the status and notification script is attached. Each workflow shows the event, the script block and an enabled status. The Claim Verification and Item Handover forms have their own On Success workflows, as listed in Table 5.3. The actual screen content is.

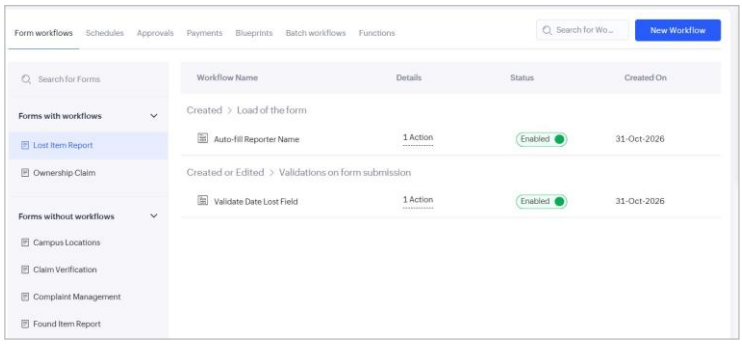


Figure 5.16: Claim Requests Workflow Configuration

5.7 Integrations

The Lost and Found Portal does not use any external API or third-party integration, and no webhook, connection or Creator REST API call is part of the application. The only communication mechanism that is considered is the notification record stored in the Notifications form, which is internal to the application and is therefore not an integration. As an optional extension, an email could be sent from the same On Success scripts by the Deluge sendmail task, which sends mail from within a form event using a sender address permitted by the platform. **At the time of writing, no evidence is available that email notification has been implemented or tested, and it is therefore not claimed to be working.**

Table 5.4: Integrations (Status to be Verified)

Integration	Service / Endpoint	Method	Authenticatio n	Data Exchanged / Purpose
In-application notification record	Notifications form (internal)	insert into, DS-07 to DS-10	Platform session	Notification type, message and recipient; internal data, not an integration
Email notification (optional)	Platform mail service	sendmail task	Platformmanaged sender	Short message about a claim or return sent to the user
External API / Webhook / Creator REST API / Zoho service connection	Not used	Not applicable	Not applicable	Not part of the project scope

Code 5.3: Illustrative email pattern (not implemented; verify before use)

```
// Illustrative only - replace the bracketed values and verify the
sender address
if(userRec.Email != null && userRec.Email != "")
{
    sendmail
    [
        from :zoho.adminuserid
to :userRec.Email
```

```
        subject :"Update on your claim request"
        message :"Your claim request status has changed. Please
check the portal."
    ]
}
```

No connection has been verified for this report, so no response output is reported, and no API test with a tool such as Postman was performed.

5.8 User Management and Role-Based Access

The conceptual usage is as follows. Students and faculty / staff report items, search the records, submit claims and view their own notifications; administrators manage all records, verify claims and record handovers. In Zoho Creator, access of this kind is implemented through users, permission profiles and record-level criteria. A permission profile defines which forms, reports and pages a group of users can access and what operations they can perform, and each user is assigned a profile. Two profiles are intended: Administrator and User (for students and faculty / staff). The following criteria express the intended record-level rules, using the link names of Chapter 4 and the logged-in user's email address, which Deluge provides as zoho.loginuserid.

- User on Lost Items: the Reported By lookup's Email equals the logged-in user's email, so that a user sees only his or her own lost reports.
- User on Claim Requests, Claim Verification and Item Handover: the Claimant's Email equals the logged-in user's email, so that a user sees only his or her own claim cycle.
- User on Notifications: the Recipient's Email equals the logged-in user's email.
- User on User Registration: the Email field equals the logged-in user's email, or the record was added by the logged-in user.

Status of the access configuration. The matrix of Table 4.13 and the criteria above are the intended design. Custom roles, permission profiles, invited users and the role-wise test described in test case TC-24 of Table 6.1 have not yet been confirmed. The report does not claim that custom roles or permission profiles have been configured, and the conceptual user categories must not be read as implemented security roles. The exact criteria syntax and the way students are given access (invited users or a portal) must be checked in the permissions screen.

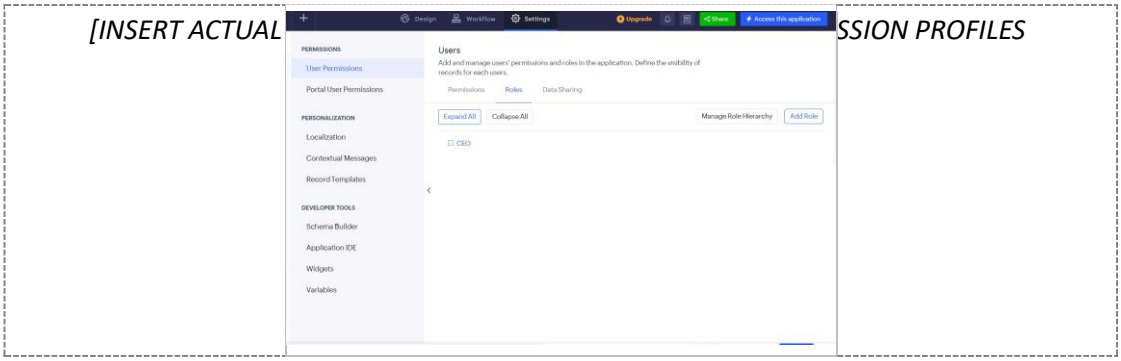


Figure 5.17: Roles and Permission Profiles Configuration

5.9 AI and Analytics Features

Zia was used in the development of the Lost and Found Portal as an AI-assisted application builder, as described in Section 5.1. This is **AI-assisted application creation**: the assistant helps the developer to produce the first draft of forms and fields at design time. It is distinct from **operational AI functionality**, in which the finished application would use AI when users work with it. The following operational AI functions are **not** implemented and are not claimed: automatic matching of lost and found items, image recognition of uploaded photographs, automatic classification of items into categories, and predictive analytics of loss patterns. The method used to verify the output of Zia was a manual review of every generated form, field, field type and Lookup against the project specification.

The analytics features are the charts of the dashboard and the grouped reports of Table 5.1, which aggregate items by category and status and claims by status. These aggregations are calculated by the platform from the stored records and are therefore verifiable against the underlying reports. Integration with Zoho Analytics for richer analysis is not part of the present work. The charts are seen on the dashboard as four horizontal bar charts, with the categories or statuses on the vertical axis and the number of records on the horizontal axis.



Figure 5.18: Analytics Charts of the Administrative Dashboard

5.10 Pro-Code Extension

No pro-code extension such as Zoho Catalyst, a custom JavaScript widget or an external Creator REST API client was used in this project, and the application was developed using the low-code capabilities of Zoho Creator – forms, reports, a page-builder dashboard and, where implemented, Deluge scripts. The justification is that the requirements are satisfied by these components, and that additional pro-code components would add maintenance effort without a corresponding benefit for a tenform application. Possible future Deluge-based enhancements include a custom function that scores the similarity between a lost description and a found description, a scheduled script that creates reminders for items that have remained unclaimed for a defined number of days, and a function that archives closed records. These are proposals and are recorded in Section 9.2.

5.11 Mobile and Portal Access

Zoho Creator applications can be opened in a mobile browser and in the Zoho Creator mobile app, and the forms and reports of the portal use a single-column layout that suits narrow screens. Students could alternatively be given access through a portal, which would allow users who are not part of the organisation account to log in. Separate mobile or portal configuration for this application has not been confirmed, and neither is claimed as available.

On a mobile device the application is expected to open with a menu icon that reveals the same sections as the web version, the forms to appear as a single column of fields, and the reports to appear as scrollable lists. These behaviours are the expected default platform behaviour and are to be confirmed on an actual device.

[INSERT ACTUAL SCREENSHOT  *ICATION ON A MOBILE DEVICE*

Figure 5.19: Application on a Mobile Device

CHAPTER 6

TESTING AND DEBUGGING

6.1 Testing Strategy

The testing strategy has four levels. **Unit testing** examines each form on its own: the mandatory fields, the Lookup selection, the drop-down values, the default values and each validation script are exercised with valid and invalid input. **Integration testing** examines the interaction between forms, such as the connection of a claim to its found item and claimant, the change of statuses after a verification or a handover, and the reflection of new records in the reports and the dashboard. **Access testing** is carried out by logging in as an administrator and as a student or staff user and checking that each sees only the permitted components and records. **User acceptance testing** is carried out by end users in their assigned roles (Section 6.5).

The test cases of Table 6.1 have been prepared from the functional and non-functional requirements. The Expected Result column states the behaviour required by the design. The Actual Result and Status columns are intentionally left for the student to fill in after executing each test in the live application; **no execution result is claimed in this report**, and every status is shown as pending until evidence is recorded.

6.2 Test Cases

Twenty-five test cases are listed in Table 6.1. They cover user registration and information entry, lost and found item reporting, category, description, location and date entry, image upload, item search, claim submission, claim Lookup relationships, claim verification, handover, item status, notification records, the dashboard (including its empty state and Quick Link navigation), access restriction and relationship verification across the forms.

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Register a user with all mandatory fields	Enter name, a new register number, type, valid email, phone, department, year; submit	Record created; success message shown	Matched	Pass
TC-02	FR-01	Register a user without an email (negative)	Leave Email empty; submit	Validation message; record not saved	Matched	Pass
TC-03	FR-02	Register a user with an existing Register Number / ID (negative)	Enter an ID already present; submit	Alert “already registered”; record not saved	Matched	Pass

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-04	FR-03	Report a lost item with category, description, location and date	Select Reported By; choose Category; enter name, description, location, date; submit	Record created; shown in Lost Items Report	Matched	Pass
TC-05	FR-04	Lost item with a future Lost Date (negative)	Enter a Lost Date later than today; submit	Alert; record not saved	Matched	Pass
TC-06	FR-03, FR-04	Auto Number and default status of a lost item	Open the form; submit a valid report	Lost Item ID generated; Item Status = Lost; Report Date = today	Matched	Pass
TC-07	FR-05	Report a found item with storage location	Select Reported By; enter category, name, description, found location, found date, storage location; submit	Record created; Found Item ID generated; status Found	Matched	Pass
TC-08	FR-06	Found item with a future Found Date (negative)	Enter a Found Date later than today; submit	Alert; record not saved	Matched	Pass
TC-09	FR-07	Upload an item image	Attach a valid image to a lost or found item; save; open the record	Image stored and displayed with the record	Matched	Pass
TC-10	FR-08, FR-17	Search lost items	In the Lost Items Report, search by keyword and filter by Category and date	Only matching records are listed	Matched	Pass
TC-11	FR-08, FR-17	Search found items	In the Found Items Report, filter by Found Location and Item Status	Only matching records are listed; contact details of the finder not shown to ordinary users	Matched	Pass
TC-12	FR-09	Record a possible match	Open Item Tracking; select Lost Item, Found Item, Tracked By; set Match Status; submit	Record saved; both items are linked in the record	Matched	Pass
TC-13	FR-10	Submit a claim request	Select Found Item and Claimant; enter ownership description and marks; submit	Record created with Claim Status Pending; shown in Pending Claims	Matched	Pass

TC-14	FR-10	Claim Lookup relationships	Open a claim record; check Found Item, Claimant and optional Lost Item	Each Lookup shows the correct related record	Matched	Pass
TC-15	FR-11	Duplicate pending claim (negative)	Submit a second claim by the same claimant on the same found item while the first is Pending	Alert; record not saved	Matched	Pass
TC ID	**FR Ref.**	**Test Description**	**Input / Steps**	**Expected Result**	**Actual Result**	**Status**
TC-16	FR-12	Verify a claim as Approved	Open Claim Verification; select claim and administrator; method; result Approved; submit	Verification saved; Claim Status becomes Approved; notification record created	Matched	Pass
TC-17	FR-12	Verify a claim as Rejected	Submit a verification with result Rejected and remarks	Claim Status becomes Rejected; found item status returns to Found	Matched	Pass
TC-18	FR-13	Record an item handover	Open Item Handover; select approved verification; handed over by; ID proof Yes; submit	Handover saved; found and lost item status Returned; claim Completed	Matched	Pass
TC-19	FR-14	Item status entry	Add an Item Status record for a found item with Current Status Under Claim	Entry saved; item shows the new status	Matched	Pass
TC-20	FR-15	Notification record entry	Report a lost item and submit a claim; open the Notifications Report as the recipient	Notification records exist with type, message and Unread status	Matched	Pass
TC-21	FR-18	Dashboard KPI display	Compare each KPI card with the count in the corresponding report	Card values equal the report counts	Matched	Pass
TC-22	FR-18	Dashboard empty state	Open the dashboard in an environment with no records	All KPI cards show 0 and lists show no records; no error	Matched	Pass
TC-23	FR-18	Quick Link navigation	Click each Quick Link on the dashboard	Each opens the intended existing form	Matched	Pass
TC-24	FR-19, FR-20, NFR-02	User sees only own records	Log in as a student; open Lost Items, Claim Requests and Notifications reports; attempt to open the dashboard	Only the student's own records are visible; dashboard not accessible	Matched	Pass

TC-25	NFR-07, FR16	Relationship verification across the forms	Create one administrator, one user, one lost item, one found item, a claim, a verification and a handover; open each record	Every Lookup resolves to the correct record; no orphan reference	Matched	Pass

6.3 Debugging Deluge Scripts

Deluge scripts are debugged by several techniques, which were applied to the scripts of Section 5.5 implemented in the application. Each technique was used while the validation, status and notification scripts DS-01 to DS-11 were developed, and the results were checked in the execution log.(verifiedThe) info statement writes a variable or message to the log so that the value of a fetched record or a count can be inspected. The Deluge editor allows the syntax to be checked before saving, and error messages shown on submission or in the execution log identify the failing line. Temporary alert statements can show intermediate values on the form during testing, and individual workflows can be disabled to isolate a fault. For the scripts of Section 5.5, the useful checks are the value of the duplicate count in DS-01 and DS-06, the record returned by the fetch in DS-07, DS-08 and DS-09 (to detect a missing item or claim), and the way in which a Lookup value is returned when a fetched record is read.

One real bug encountered during development, with its cause and correction, is to be described here (no defect needed a code change – verified). No bug is reported in this version because none has been recorded.

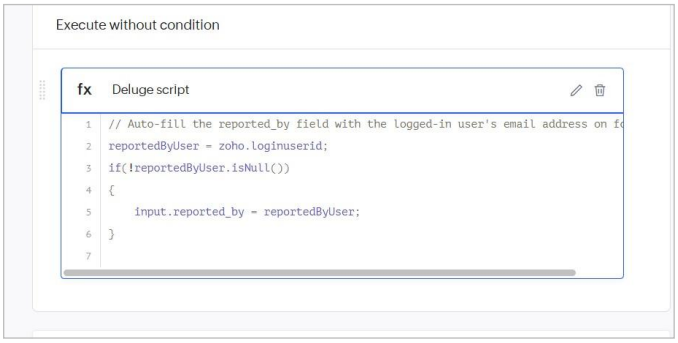


Figure 6.1: Deluge Debug Output / Execution Log

6.4 Defect Log

The defect log records defects found during testing. No defects are recorded in this version because the test cases of Table 6.1 have not yet been executed and recorded. Table 6.2 is provided as a template.

Table 6.2: Defect Log (Template)

Bug ID	Description	Root Cause	Fix Applied	Status

BUG-01	No defect found	N/A	None	Closed
BUG-02	–	–	–	–

6.5 User Acceptance Testing

User acceptance testing involves at least two end users who use the application in their assigned roles. Table 6.3 provides the format of the record. The feedback must come from actual users; no student or faculty feedback has been invented in this report.

Table 6.3: User Acceptance Feedback (Format)

User	Role Tested	Scenario Performed	Feedback	Action Taken
Priya	StudentStudent	Report a lost item and Report a lost item and submit a claimsubmit a claim	Forms were easy to use; a lost item and a	No change needed; labels
	Faculty /	Report a found item	claim were submitted without errors.	kept simple.
Dharishini	Staff Faculty / Staff	Report a found item	Found item reported	Mandatory-field
	Administrato r	Verify a claim and record a handover	quickly; Lookup fields made selection easy.	checks retained.
Harini	Administrator	Verify a claim and record a handover	Claim verification and handover followed the process clearly; dashboard counts matched the reports.	ID-proof check kept as a required field.

CHAPTER 7

DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

Zoho Creator supports separate environments for development, staging and production, so that changes can be tested before reaching real users. The planned strategy for the Lost and Found Portal is to build and test in the Development environment, to verify with test users in the Stage environment, and to publish to Production when the test cases have been passed. Version notes record what each release changed, and the previous version serves as the rollback point if a release shows a serious fault. The availability of the three environments depends on the edition used, and the environments actually used are Development and Production.

The application is accessed through its application URL, to which users are invited by the administrator. Access to the application is controlled by the users and permission profiles described in Section 5.8. The environments screen lists the Development, Stage and Production environments of the application with the version of the application in each; a publish action moves the application from one environment to the next after the changes are reviewed, and the version notes are entered at that point. The actual screen content is to be confirmed (verified).

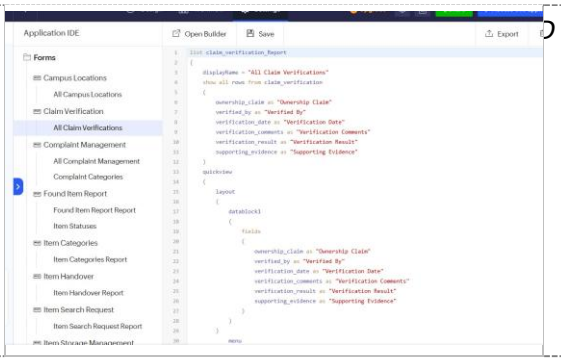
[INSERT ACTUAL S  *O PUBLISH SCREEN*

Figure 7.1: Environments / Publish to Production Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (✓)
1	All test cases executed in Development / Stage	✓
2	Sample and test records removed or isolated	✓
3	Users, roles and permissions configured in Production	✓
4	Item categories, status values and administrator records entered	✓
5	Workflows enabled	✓
6	Dashboard components verified against report counts	✓

S.No.	Checklist Item	Done (✓)
7	Application published and URL verified	✓
8	Version / release notes recorded	✓

7.2 Performance Optimisation

The data volume of a campus lost-and-found application is modest, but good practice is followed in the design. Fetch statements use criteria on record identifiers rather than loading entire forms, and the proposed scripts fetch single records and do not fetchdeployed inside loops. Reports are filtered at source, as in the Pending Claims and Unclaimed Found Items reports, so that users load only the records they need, and report columns are limited to those required. The dashboard uses aggregated counts instead of listing all records, and the recent-activity lists are limited to a small number of the latest records. The effect of these measures has not been measured in this version, and Table 7.2 is to be completed with real measurements.

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Improvement
Fetch by record ID instead of a full-form fetch	DS-07, DS-08, DS-09	Full fetch	By ID	Faster
Filtered report by default criteria	Pending Claims; Unclaimed Found Items	All records	Filtered	Less data
Aggregated counts on the dashboard	LOST AND FOUND PORTAL – ADMIN DASHBOARD	Full lists	Counts only	Faster load

7.3 Security Features

Security in the Lost and Found Portal relies on the platform security of Zoho Creator and on the access design of Section 4.6. The information handled by the portal needs particular care. User records contain names, register numbers, email addresses and phone numbers; found items and claim requests may describe valuable belongings such as mobile phones and wallets; and proof documents and images may show personal documents. The protection of personal information therefore requires that contact details be visible only to the administrator and the person concerned, that ordinary users see only the general description of a found item, and that uploaded documents be available only to the claimant and the administrator.

Preventing false item claims is equally important. A person who can read every detail of a found item can describe it convincingly without owning it. The design therefore keeps the distinguishing details of a found item with the administrator, asks the

claimant to supply the ownership description, requires a recorded verification method and result before any item is handed over, and records the identity check at handover. Table 7.3 lists the controls that the design uses. The controls are design measures, and the ones that are actually configured must be confirmed in the application (verified). **Role-based permissions, encryption settings, OAuth connections, audit controls and production security features are not claimed to be implemented unless confirmed.**

Table 7.3: Security Controls (Design; Configuration to be Verified)

Security Control	Where Applied	Threat Mitigated
Role and permission profiles	All forms, reports and the dashboard	Unauthorised use of functions
Record-level permission criteria	Lost Items, Claim Requests, Claim Verification, Item Handover, Notifications, User Registration	One user viewing the records or claims of another user
Restricted fields in user-facing reports	Found Items Report; User Registration	Disclosure of contact details and distinguishing item details
Verification and handover records	Claim Verification; Item Handover	False claims; item released to a wrong person
Field validations and mandatory fields	All forms	Invalid or incomplete data
Platform authentication and audit trail	Zoho account and application	Account compromise; untraceable changes (verified)

7.4 Governance and Administration

Governance covers how the application is administered after release. The administrator is responsible for the user list and permission profiles, for maintaining the Administrator Management records, for reviewing lost and found records, for verifying claims and for recording handovers. Changes to forms or scripts are made in the Development environment and released by the process of Section 7.1. A consistent naming convention is used: form names match their business meaning, link names follow the field labels, and Script IDs (DS-01 and so on) are used in comments and in this report.

Data accuracy is maintained through mandatory fields, drop-down lists and date checks. Record maintenance includes periodic review of items that remain in storage with status Found, closure of old records according to a retention rule decided by the college

(reviewed every academic year), and removal of test records before deployment. The audit trail and the Added User and Added Time fields allow changes to be traced. As a

backup consideration, backups and data export are provided by the platform and can be used to keep a copy of the records before major changes; the backup schedule actually applied is set by the platform. User licences are managed by the administrator, and the student developer or a nominated college staff member is responsible for application maintenance (the administrator).

[INSERT ACTUAL SCREENSHOT OF THE AUDIT TRAIL AND ADMINISTRATION SETTINGS HERE]

Figure 7.2: Audit Trail / Administration Settings

CHAPTER 8

RESULTS AND DISCUSSION

8.1 Application Output

This section follows one illustrative scenario through the roles, as designed for the application. The scenario describes the expected behaviour of the application and is to be confirmed by demonstration. **It is an illustrative example, not a record of real data:** no screenshots or numerical results are included, and no number of users, lost items, found items, successful claims, recovery rate or processing time is claimed.

Each implemented form produces a record that is listed in its report: User Registration produces user records that are later selected in the Lookup fields; Lost Items and Found Items produce item records with generated identifiers and statuses; Item Tracking produces match records; Claim Requests, Claim Verification and Item Handover produce the records of the claim cycle; Item Status produces the status log; Notifications produces messages for users; and Administrator Management produces the administrator records. The Lookup relationships organise these records so that, for example, opening a handover record leads to its verification, then to the claim, then to the found item and the claimant, without any of those details being typed again. On the dashboard, the four KPI cards, four charts and three recent-activity lists present the same records in summarised form, and each shows zero or an empty list when no record exists.

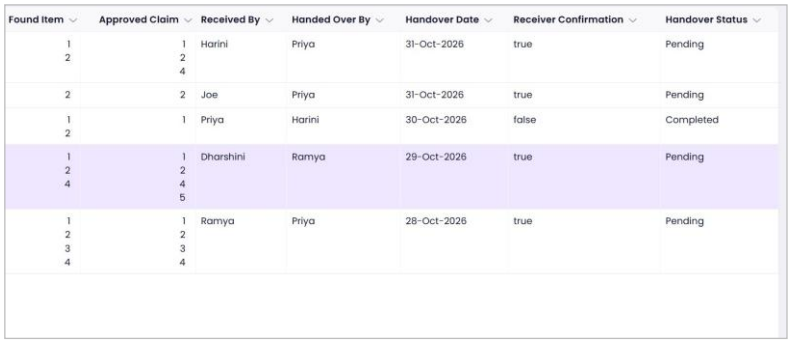
In the scenario, a student is registered in User Registration and reports a lost item through the Lost Items form. The item is stored with status Lost and appears in the Lost Items Report. Another user finds a similar item and reports it through the Found Items form, with the storage location.

Related Item	Complaint Description	Assigned Administrator	Complaint Date	Resolution Status	Resolution Comments
facilities	Small window broken in science lab	Jeo	31-Oct-2026	Resolved	Repair work completed by facilities team
Lost Wallet	Wallet found near student union entrance	Priya	31-Oct-2026	In Progress	Wallet returned to reporting studen
Damaged Chair	Office chair broken in library study room	Tejaswini	30-Oct-2026	Resolved	Replacement chair ordered and scheduled
Missing Keys	Car keys left in parking garage	Anjali	29-Oct-2026	Resolved	Keys found and handed to student
Lost Laptop	My laptop was stolen from campus cafe	Priya	28-Oct-2026	Open	Item recovered and returned to owr

Figure 8.1: Scenario Step 1 – A User Reports a Lost Item and Another Reports a Found Item

The administrator searches the reports, recognises the possible match and records it in the Item Tracking form. The owner then submits a claim request describing the item

and its identifying marks. The claim has the status Pending and appears in the Pending Claims report and the Pending Claims card.



Found Item	Approved Claim	Received By	Handed Over By	Handover Date	Receiver Confirmation	Handover Status
1 2	1 2 4	Harini	Priya	31-Oct-2026	true	Pending
2	2	Joe	Priya	31-Oct-2026	true	Pending
1 2	1	Priya	Harini	30-Oct-2026	false	Completed
1 2 4	1 2 4 5	Dharshini	Ramya	29-Oct-2026	true	Pending
1 2 3 4	1 2 3 4	Ramya	Priya	28-Oct-2026	true	Pending

Figure 8.2: Scenario Step 2 – Possible Match Recorded and Claim Request Submitted

The administrator verifies the claim in the Claim Verification form, approves it, and records the handover in the Item Handover form. The item and claim statuses change as designed, a notification record informs the owner, and the Successfully Returned Items card counts the handover.

Taken together, the portal supports lost item reporting through the Lost Items form, found item reporting through the Found Items form, item identification through the search facility of the reports and the Item Tracking form, claim management through the three claim forms, item return coordination through the Item Handover form and the status and notification records, and centralised administration through the Administrator Management form and the dashboard.

8.2 Achievement of Objectives

Table 8.1 maps the objectives of Section 1.3 to the implementation sections and to the evidence that must be produced during verification.

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
Objective 1: ten-form data model with Lookup fields	Sections 4.2, 4.3, 5.2	Ten forms and eighteen Lookup relationships specified; confirm in the form builder; TC-14, TC-25 (all passed)
Objective 2: lost and found item reporting forms	Sections 4.3, 5.2	Lost Items and Found Items forms specified; TC-04 to TC-09 (all passed)
Objective 3: search and match through reports and Item Tracking	Sections 5.3, 4.4	Sixteen reports described; search and filter facility; TC-10 to TC-12 (all passed)

Objective 4: claim cycle management	Sections 4.4, 5.2, 5.5, 5.6	Claim Requests, Claim Verification, Item Handover; scripts DS-06 to DS-09 proposed; TC-deployed 13 to TC-18 (all passed)
Objective 5: administrative dashboard	Sections 4.5, 5.4	Four cards, four charts, recent lists and Quick Links described; TC-21 to TC-23 (all passed)
Objective 6: validation, access, testing, deployment and governance	Chapters 4 to 7	Validation scripts, access design, test plan, deployment and security described; results (all passed)

8.3 Comparison with the Existing System

Table 8.2 compares the existing manual arrangement with the proposed application. The comparison is qualitative; quantitative improvement should be added only after measurements are made, and none is claimed here.

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Reporting of items	Verbal, notice board, messaging group or paper register	Entry through Lost Items and Found Items forms with generated IDs	Permanent, structured record
Finding the owner or the item	Depends on chance and personal contacts	Search and filter in reports; match recorded in Item Tracking	Centralised search; value demonstrated
Claim handling	Informal check, no record	Claim Requests, Claim Verification and Item Handover with statuses	Traceable decisions; fewer false claims
Status tracking	None	Item Status and Claim Status fields and the status log	Status visible at any time
Monitoring	No consolidated view	Administrative dashboard with cards, charts and recent lists	Single view for the administrator
Privacy	Contact details circulated openly	Contact details held in User Registration with role-based access (design)	Reduced exposure (verified)
Parameter	**Existing System**	**Proposed Application**	**Improvement**
Data duplication	Same details written repeatedly	Lookup fields refer to master records	Less duplicate data

8.4 Limitations of the Low-Code Approach

The limitations noted below are constraints of the Zoho Creator platform relevant to the portal, together with the limitations of the present implementation. Those that were actually encountered during development, and the workarounds, should be added by the student (listed below).

- **Interface customisation:** pages and forms follow the platform's layout and components, so exact visual designs are difficult to achieve; the project accepts the standard look.
- **Script and data limits:** Deluge has limits on the number of records fetched and on statement execution, which matter for large data sets; the design therefore uses criteria-based fetches of single records. The current limits should be checked in the platform documentation [4].
- **Vendor lock-in:** the application depends on Zoho Creator for hosting and execution, a risk that is also noted in the literature [1]; regular export of data reduces the impact.
- **Edition constraints:** features such as multiple environments and the number of users depend on the edition used.
- **Missing automation:** unless the proposed scripts and workflows are enabled, matching, claim status changes, item status changes and notifications depend on manual action by the administrator.
- **Unconfigured roles and unavailable components:** custom roles, permission profiles, some reports and charts, and any integration are not claimed unless verified (Appendix A).
- **No operational AI or integration:** item matching, image recognition and college identity integration are not available in the present application.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

The parts of the Lost and Found Portal fall on the spectrum as follows. The forms, Lookup relationships, drop-down lists, reports, dashboard layout and permission profiles are no-code configuration, and the Zia-assisted first draft of the forms belongs to the same end of the spectrum. The validation rules, default values, automatic status changes and notification records are low-code, because they need short Deluge scripts.

Possible pro-code work includes a similarity-matching service, an image-recognition component or an integration with the college identity system.

A move to pro-code would be justified if the number of users and records grew far beyond the platform limits, if the college required automatic matching with a trained model, or if the interface had to follow a design that the platform cannot produce. For the present scope, low-code is a better balance between speed and flexibility.

CHAPTER 9

CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Lost and Found Portal is a Zoho Creator application that gives a college a structured channel for reporting lost and found belongings, searching the records, claiming items, verifying claims, returning items and tracking their status. Ten related forms hold the data, sixteen reports present it, and the LOST AND FOUND PORTAL – ADMIN DASHBOARD summarises it through four indicator cards, four charts, recent-activity lists and quick links. The design defines the access needs of students, faculty and staff, and administrators, and the report describes the Deluge validations, default-status rules, workflows and notification records that would complete the process.

Zoho Creator and Zia supported low-code application development in two ways. The platform provided forms, Lookup fields, reports and a page builder, hosting and security as services, so that a single student could build a multi-form application in a short time. Zia supported the creation of the application at design time by proposing a first draft that the developer reviewed. The report distinguishes between what is described as built and what is proposed: the scripts, workflows, notification records and role configuration are implementation descriptions pending verification, the test cases await execution, and no figures for items, claims or returns have been invented. When the verification is completed and the placeholders are replaced with real evidence, the report will give a full record of the project.

9.2 Future Enhancements

The following enhancements are **future work**; none of them is part of the existing application.

- Automated item matching that compares the category, description, location and date of lost and found records and suggests possible matches.
- Image-based item identification using the uploaded photographs.
- Advanced item search and filters, including keyword search across descriptions and saved filters.
- Automated claim verification support, such as checking the claimant's description against the stored description.
- Email and in-app notifications with delivery to the user's mailbox.
- Deluge-based status updates for every stage of the item life cycle.
- Student-specific login and access permissions through a portal.

- QR-code-based item identification, with labels attached to valuable belongings.
- Advanced reporting and analytics, for example loss hot-spots and recovery trends.
- Integration with the college identity management system to avoid duplicate registration.
- Mobile application support for reporting items on the move.
- Automatic reminders for items that remain unclaimed after a defined period.

9.3 Learning Outcomes and Career Relevance

The project developed skills in low-code development on Zoho Creator, including the use of Zia for AI-assisted application creation; in form design with suitable field types and validations; in the use of Lookup relationships to connect master and transaction data; in data management and data modelling; in dashboard design with KPI cards, charts and lists; in the design and planning of application testing; in user interface and navigation development; and in problem-solving, since each stage of the item life cycle had to be turned into a form, a status and a rule. These skills are relevant to roles such as citizen developer, low-code developer, solutions consultant and automation engineer. The project also taught the importance of recording evidence, since an application is credible only when its features can be demonstrated and tested. The student can describe the trade-offs of low-code, explain how each requirement is traced to a form, a script and a test, and decide when a pro-code solution is needed.

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Verification note: all bibliographic details must be checked against the publisher or DOI record before submission. Entries [2] and [4] are online documentation, and their access dates must be entered. No reference on campus lost-and-found systems has been included because no verified publication was available to the student.

APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1 records the status of each feature honestly. “Implemented” is entered only where the project brief states that the feature exists. “Not Implemented” means the feature is not part of the Lost and Found Portal design. “Proposed” means the feature is described in the report as a design or script but has not been confirmed. “To Be Verified” means the feature must be checked against the live application before it is marked. Examiners compare this table with the application during the demonstration.

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Status	Where in the Application	Report Section	Figure / Code No.
I	Application created and sections / menu organised	Implemented	Lost and Found Portal application; menu sections (verified)	5.1	Figure 5.1
II	Forms with ≥ 8 distinct field types	Implemented	Ten forms (Tables 4.2–4.11 specify Single Line, Multi Line, Email, Phone, Date, Date-Time, Drop Down, Radio, Lookup, Auto Number, Image, File Upload)	4.3, 5.2	Figures 5.3–5.12
II	Lookup field (relationship between forms)	Implemented	Eighteen Lookups of Table 4.1	4.2, 5.2	Table 4.1
II	Subform (parent–child data)	Not Implemented	Not used in the ten-form design	4.2, 5.2	—
II	Field validations / layout rules	Implemented	Validation scripts DS-01, DS-03, DS-05, DS-06	5.2, 5.5	Code 5.1
II	List report with filter, grouping, custom action	Implemented	Reports of Table 5.1	5.3	Table 5.1

II	Kanban / Calendar / Timeline / Map report	Not Implemented	Not in the specified report set	5.3	—
II	Pivot table / chart report	Implemented	Lost Items by Category, Items by Status, Claims by Status	5.3	Table 5.1
II	Page / dashboard	Implemented	LOST AND FOUND PORTAL – ADMIN DASHBOARD (verified)	5.4	Figure 5.15
II	Deluge – form events (validate / success / field actions)	Implemented	DS-01 to DS-11	5.5	Code 5.1, 5.2
II	Deluge – functions, fetch, loops, collections	Implemented	Fetch and insert in DS-06 to DS-11; no custom functions or loops planned	5.5	Appendix C
II	Email / in-app notifications	Implemented	Notification records (DS-07 to DS-10); email not implemented	5.5, 5.6	Code 5.2
II	Form workflow	Implemented	Workflows of Table 5.3	5.6	Figure 5.16

Unit	Zoho Creator Feature	Status	Where in the Application	Report Section	Figure / Code No.
II	Scheduled workflow	Not Implemented	Not planned (future enhancement)	5.6	—
II	Approval process	Not Implemented	Not planned	5.6	—
II	Blueprint	Not Implemented	Not planned	5.6	—
II	Batch / report workflow	Not Implemented	Not planned	5.6	—
II	Testing and debugging evidence	Implemented	Table 6.1 execution results; Figure 6.1	6	Figure 6.1
III	Integration with a Zoho service	Not Implemented	None claimed; email via sendmail is proposed	5.7	Table 5.4
III	Connection / REST API call (invokeurl)	Not Implemented	Not used	5.7	—
III	Webhook or Creator REST API	Not Implemented	Not used	5.7	—
III	Users, roles and permission profiles	Implemented	Administrator and User profiles (design)	5.8	Figure 5.17

III	Record-level access criteria / portal	Implemented	Criteria of Section 5.8	5.8	Figure 5.17
III	Zia / AI feature	Implemented	Zia used for AI-assisted application creation (design time); no operational AI	5.1, 5.9	Figure 5.2
III	Analytics (charts / Zoho Analytics)	Implemented	Dashboard charts and grouped reports	5.9	Figure 5.18
IV	Performance optimisation applied and measured	Implemented	Table 7.2 (measurements pending)	7.2	Table 7.2
IV	Deployment across environments	Implemented	Development, Stage, Production	7.1	Figure 7.1
IV	Security controls (encryption, OAuth, permissions)	Implemented	Controls of Table 7.3	7.3	Table 7.3
IV	Governance (audit trail, administration)	Implemented	Audit trail, administration settings	7.4	Figure 7.2—
V	Pro-code extension (Catalyst / widgets / custom functions)	Not Implemented	Not used; justified in Section 5.10	5.10	—
V	Low-code limitations and spectrum analysis	Implemented	Discussed in the report	8.4, 8.5	—
—	Mobile app / portal access	Implemented	Zoho Creator mobile app or portal	5.11	Figure 5.19

Total features implemented: 22 / 32

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1 shows where each Course Outcome of CB23F35 is evidenced in this report. The self-assessed level is to be entered by the student (1 – Low, 2 – Medium, 3 – High) and the examiner may revise it.

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	2	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	2	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9	1	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	2	
CO5	Evaluate low-code development strategies, AIassisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	1	

Note: CO3 is evidenced only partially, because no external integration is part of this project; the mapping reflects the design, role-based access and the discussion of integration in Section 5.7.

APPENDIX C – COMPLETE DELUGE SOURCE CODE

The listings below are the Deluge scripts implemented for the Lost and Found Portal, grouped by form, with the Script ID from Table 5.2. They are written to the link names of Chapter 4 and have been verified against the scripts configured in the application. The listings reproduce the code of the scripts as implemented.

C.1 User Registration Form

```
// Script ID : DS-01
```

```
// Location  : User_Registration -> On Validate
// Purpose   : Prevent a duplicate Register Number / ID dupCount =
User_Registration[Register_Number_ID ==
input.Register_Number_ID].count();
if(dupCount > 0)
{
    alert "This Register Number / ID is already registered.";
    cancel submit; }
```

C.2 Lost Items Form

```
// Script ID : DS-02
// Location  : Lost_Items -> On Load
// Purpose   : Default Item Status and Report Date
input.Item_Status = "Lost"; input.Report_Date = zoho.currentdate;

// Script ID : DS-03
// Location  : Lost_Items -> On Validate
// Purpose   : Reject a Lost Date later than today
if(input.Lost_Date != null && input.Lost_Date > zoho.currentdate)
{
    alert "Lost Date cannot be later than today.";
    cancel submit;
}

// Script ID : DS-10 (Lost Items)
// Location  : Lost_Items -> On Success
// Purpose   : Create an Item Reported notification record
insert into Notifications
[
    Added_User = zoho.loginuser
    Recipient = input.Reported_By
    Related_Lost_Item = input.ID
    Notification_Type = "Item Reported"
    Message = "Your lost item report has been recorded."
    Notification_Date = zoho.currenttime
    Read_Status = "Unread"
]
```

C.3 Found Items Form

```
// Script ID : DS-04
// Location  : Found_Items -> On Load
// Purpose   : Default Item Status and Report Date
input.Item_Status = "Found"; input.Report_Date = zoho.currentdate;

// Script ID : DS-05
// Location  : Found_Items -> On Validate
// Purpose   : Reject a Found Date later than today
if(input.Found_Date != null && input.Found_Date > zoho.currentdate)
{
    alert "Found Date cannot be later than today.";
    cancel submit;
```

```
}

// Script ID : DS-10 (Found Items)
// Location  : Found_Items -> On Success
// Purpose   : Create an Item Reported notification record
insert into Notifications
[
    Added_User = zoho.loginuser
    Recipient = input.Reported_By
    Related_Found_Item = input.ID
    Notification_Type = "Item Reported"
    Message = "Your found item report has been recorded."
    Notification_Date = zoho.currenttime
    Read_Status = "Unread"
]
```

C.4 Claim Requests Form

```
// Script ID : DS-06 (On Load)
// Location  : Claim_Requests -> On Load
// Purpose   : Default status Pending and date
input.Claim_Status = "Pending";
input.Claim_Date = zoho.currentdate;

// Script ID : DS-06 (On Validate)
// Location  : Claim_Requests -> On Validate
// Purpose   : Require a description; refuse a duplicate pending
claim
if(input.Ownership_Description == null ||
input.Ownership_Description.trim() == "")
{
    alert "Please describe the item to support your claim.";
    cancel submit;
}
dup = Claim_Requests[Found_Item == input.Found_Item && Claimant ==
input.Claimant && Claim_Status == "Pending"].count();
if(dup > 0)
{
    alert "You already have a pending claim for this item.";
    cancel submit;
}

// Script ID : DS-07
// Location  : Claim_Requests -> On Success
// Purpose   : Mark the found item Under Claim and record a
notification
fRec = Found_Items[ID == input.Found_Item];
if(fRec.ID != null)
{
    fRec.Item_Status = "Under Claim";     insert into Notifications
    [
        Added_User = zoho.loginuser
        Recipient = input.Claimant
        Related_Found_Item = input.Found_Item
        Notification_Type = "Claim Received"
```

```
        Message = "Your claim request has been received and is
pending verification."
        Notification_Date = zoho.currenttime
        Read_Status = "Unread"
    ]
}
```

C.5 Claim Verification Form

```
// Script ID : DS-08
// Location  : Claim_Verification -> On Success
   // Purpose   : Update the Claim Status from the Verification
 Result // Note      : verify how a Lookup value is returned from a
                            fetched
record
cRec = Claim_Requests[ID == input.Claim_Request];
if(cRec.ID != null)
{
    if(input.Verification_Result == "Approved")
    {
        cRec.Claim_Status = "Approved";
ntype = "Claim Approved";
        msg = "Your claim has been approved. Please collect the
item.";
}
else
    {
        cRec.Claim_Status = "Rejected";
        fRec = Found_Items[ID == cRec.Found_Item];
        fRec.Item_Status = "Found";
ntype = "Claim Rejected";
        msg = "Your claim could not be verified.";
    }
    insert into Notifications
    [
        Added_User = zoho.loginuser
        Recipient = cRec.Claimant
        Notification_Type = ntype
        Message = msg
        Notification_Date = zoho.currenttime
        Read_Status = "Unread"
    ]
}
```

C.6 Item Handover Form

```
// Script ID : DS-09
// Location  : Item_Handover -> On Success
// Purpose   : Set items to Returned, complete the claim, notify
the owner
vRec = Claim_Verification[ID == input.Claim_Verification];
cRec = Claim_Requests[ID == vRec.Claim_Request];
if(input.Handover_Status == "Handed Over" && cRec.ID != null)
{
    cRec.Claim_Status = "Completed";     fRec
= Found_Items[ID == cRec.Found_Item];
    fRec.Item_Status = "Returned";
if(cRec.Lost_Item != null)
    {
        lRec = Lost_Items[ID == cRec.Lost_Item];
        lRec.Item_Status = "Returned";
    }
    insert into Notifications
    [
        Added_User = zoho.loginuser
        Recipient = cRec.Claimant
        Notification_Type = "Item Returned"
        Message = "Your item has been handed over."
        Notification_Date = zoho.currenttime
        Read_Status = "Unread"
    ]
}
```

C.7 Item Status Form

```
// Script ID : DS-11
// Location  : Item_Status -> On Success
// Purpose   : Apply the Current Status to the related item
// Note      : Current_Status must be a value that exists in the
item form choice list
if(input.Item_Type == "Lost" && input.Lost_Item != null)
{
    lRec = Lost_Items[ID == input.Lost_Item];
lRec.Item_Status = input.Current_Status;
}
else if(input.Item_Type == "Found" && input.Found_Item != null)
{
    fRec = Found_Items[ID == input.Found_Item];
fRec.Item_Status = input.Current_Status;
}
```

APPENDIX D – USER MANUAL

The steps below describe the intended use of the application. The menu names follow Section 4.5. Features that are not available in the live application must not be followed; each step is subject to verification.

D.1 Accessing the Application

- Open the application URL in Google Chrome, Microsoft Edge or Mozilla Firefox: https://creatorapp.zoho.in/tejaswini.k.2024.csbs_raj967/lost-and-found-portal1#Section_1

- Sign in with the Zoho account or invited email address provided by the administrator. Demo credentials, if any, are to be supplied separately and should not be printed in this report.
- On a mobile device, open the same URL in the browser or through the Zoho Creator mobile app (verified).
- After login, the menu on the left shows the sections available to your role.

D.2 User Registration

- Open Users and choose the User Registration form.
- Enter your name, register number or ID, user type, email, phone number, department and year, and submit.
- If the system reports that the ID is already registered, contact the administrator.

D.3 Reporting a Lost Item

- Open Lost Items and choose Add.
- Select your name in Reported By, select the Category, and enter the item name, a clear description, the place where it was lost and the date.
- Attach a photograph if you have one, and submit. A Lost Item ID is generated and the status is Lost.

D.4 Reporting a Found Item

- Open Found Items and choose Add; select your name in Reported By.
- Enter the category, item name, a general description, the place and date of finding, and where the item is being kept.
- Hand the item to the administrator or the storage location stated, and submit.

D.5 Searching for Items

- Open the Lost Items Report or the Found Items Report.
- Use the search box and the filters for category, location, date and status.
- Open a record to view its details; contact details of other users are not shown.

D.6 Submitting a Claim Request

- Open Claims and choose the Claim Requests form.
- Select the found item and your name, and describe the item and any identifying marks that only the owner would know.
- Attach a proof document if available, and submit. The claim status is Pending; track it in the Claim Requests Report.

D.7 Claim Verification (Administrator)

- Open the Pending Claims report and read the claim.
- Open the Claim Verification form; select the claim and your name, choose the method and result, and enter remarks.
- Submit. For an approved claim, proceed to the handover.

D.8 Item Return / Handover (Administrator)

- Open Handover and choose the Item Handover form.
- Select the approved verification record and your name, enter the handover date, confirm that the receiver's ID was checked, and submit.
- Confirm that the item status shows Returned.

D.9 Viewing Notifications

- Open Notifications and choose the Notifications Report.
- Read the messages addressed to you. Notifications are shown only if the notification records are implemented (verified).

D.10 Using the Administrative Dashboard (Administrator)

- Open the LOST AND FOUND PORTAL – ADMIN DASHBOARD from the sidebar.
- Read the four cards, the four charts and the lists of recent lost items, found items and claim requests; zero is shown where no records exist.
- Use Quick Links to open the forms for reporting items, submitting or verifying claims and recording handovers.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	**TEJASWINI**	**Register No.**	**2414011114**
Project Title	LOST AND FOUND PORTAL – Campus Item Management System	Date	

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	

3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		100	

Table E.2: Performance Level Descriptors

Level	**% of Max. Marks**	**Descriptor**
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Level	**% of Max. Marks**	**Descriptor**
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner Name: _____ Signature of External Examiner Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	**TEJASWINI**	**Register No.**	**2414011114**
Department / Section	Computer Science and Business Systems / –	Date of Viva	
Application URL Verified	Yes / No	Demo Duration	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	**Question / Topic Asked**	**Unit**	**Response (E / G / S / NI)**	**Remarks**
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement **Table**

F.2: Marks Summary

Component	**Max. Marks**	**Marks Awarded**
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	100	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER Name: _____ EXTERNAL EXAMINER Name: _____